Candidate-pool opportunity and selection stability under finite validation in molecular property prediction: a repeated nested audit

Journal: Journal of Cheminformatics

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Additional file 3

Supplementary Figures S1–S29

Description: Contains the complete supplementary figure set and audit visualizations. All figure numbering and descriptions correspond to the supplementary methods and tables.

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Supplementary Figures S1–S29

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Figure S1. Dataset cleaning flow

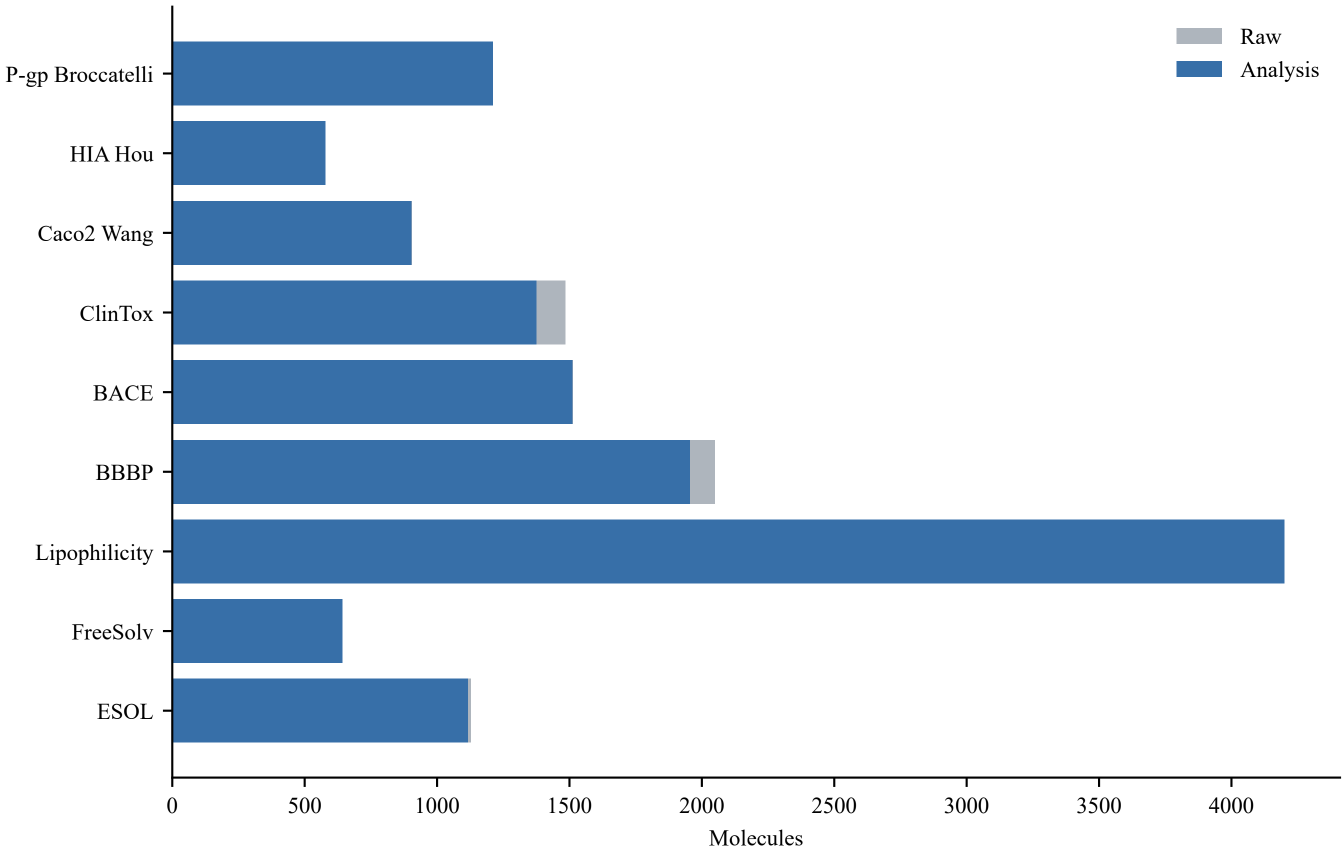


Figure S2. Candidate correlation matrices

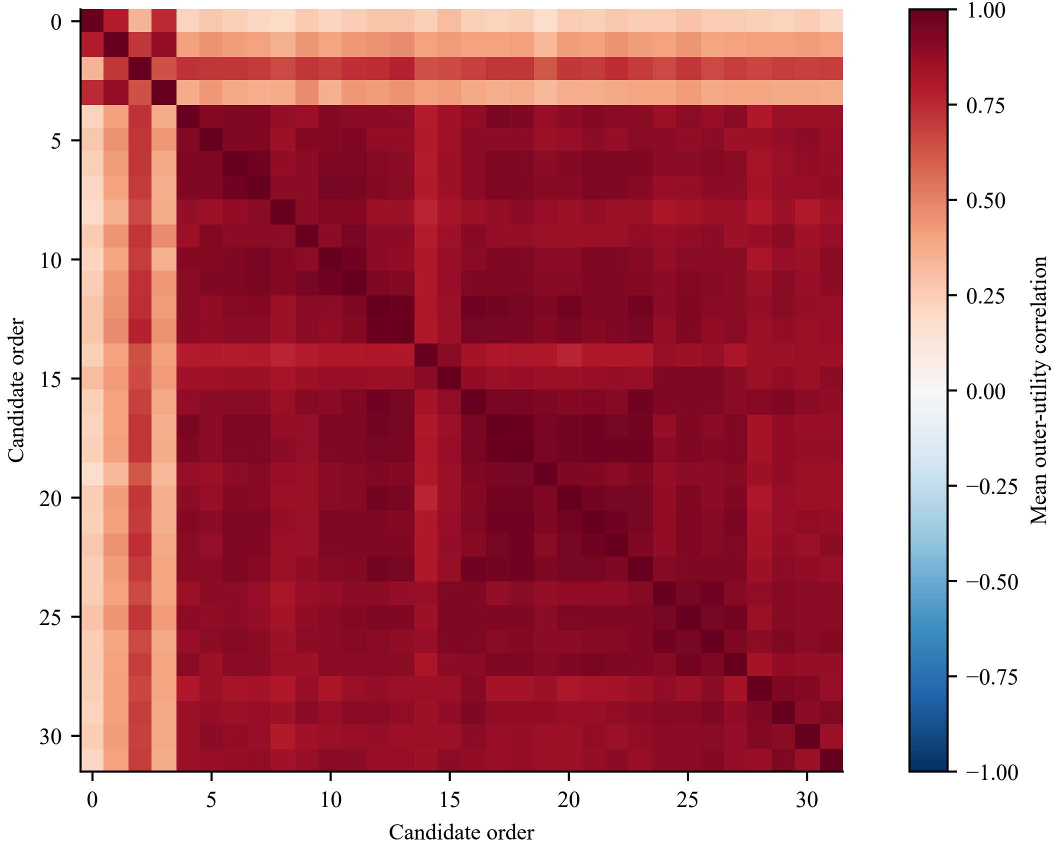


Figure S3. Eigenvalue spectra

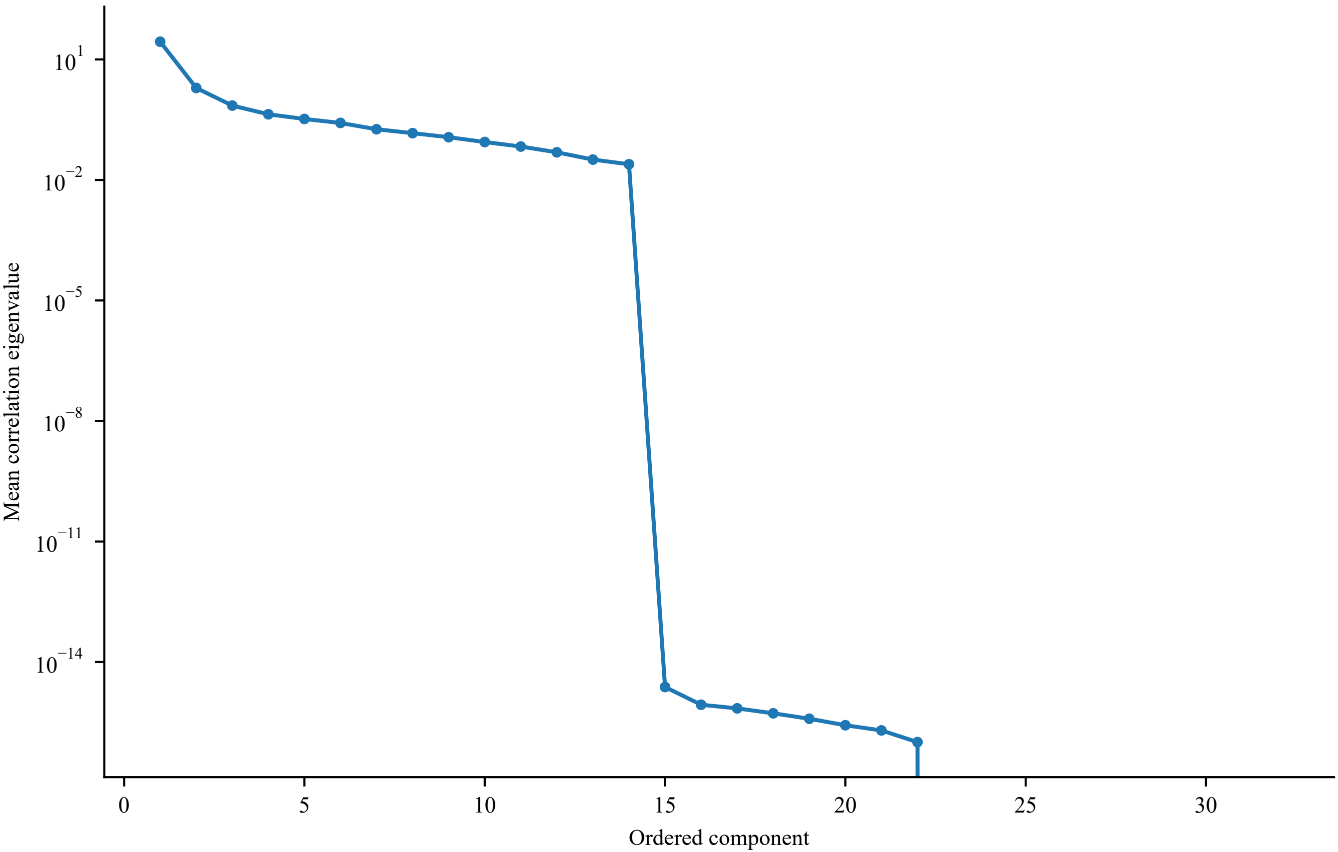


Figure S4. Raw Top-1 and raw MRR

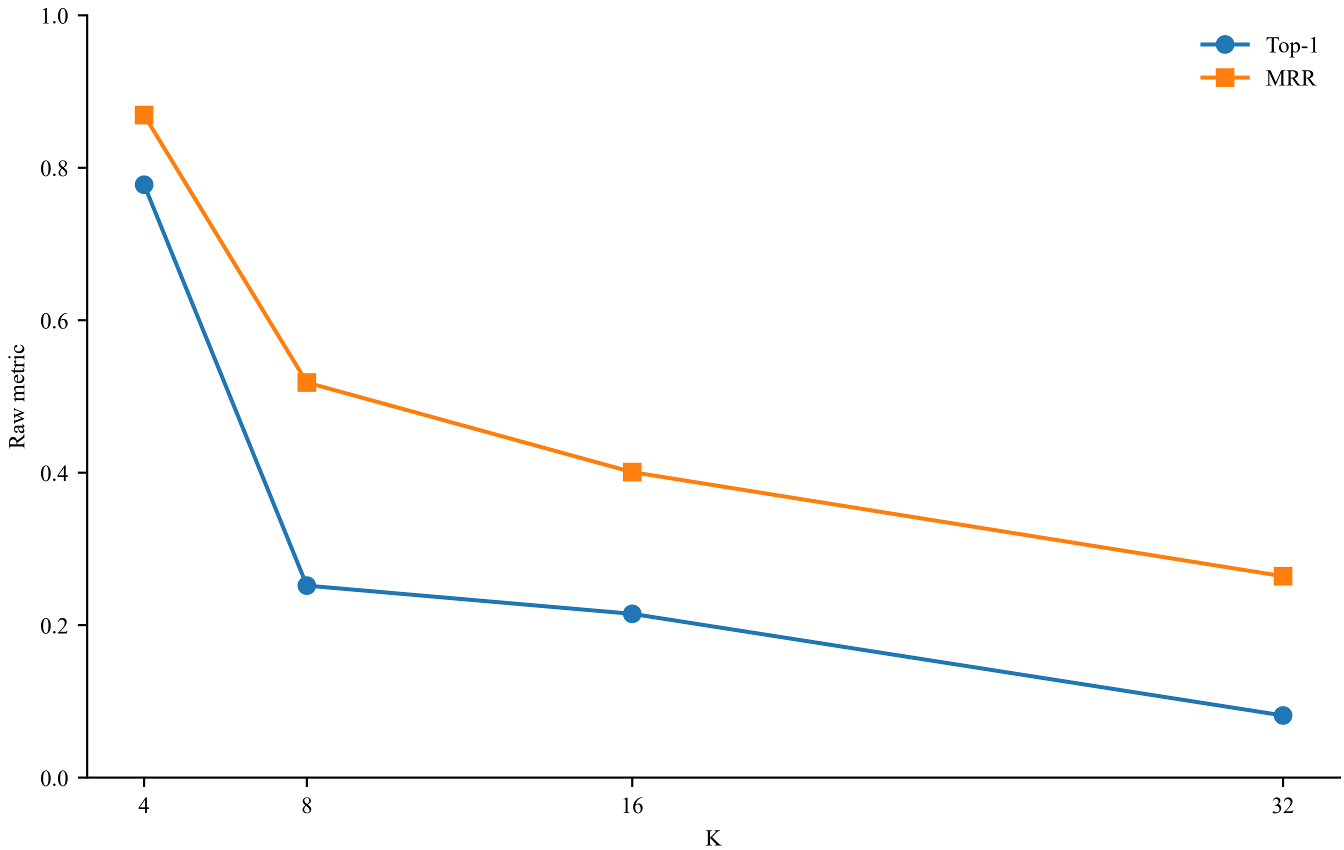


Figure S5. Random-rank permutation calibration

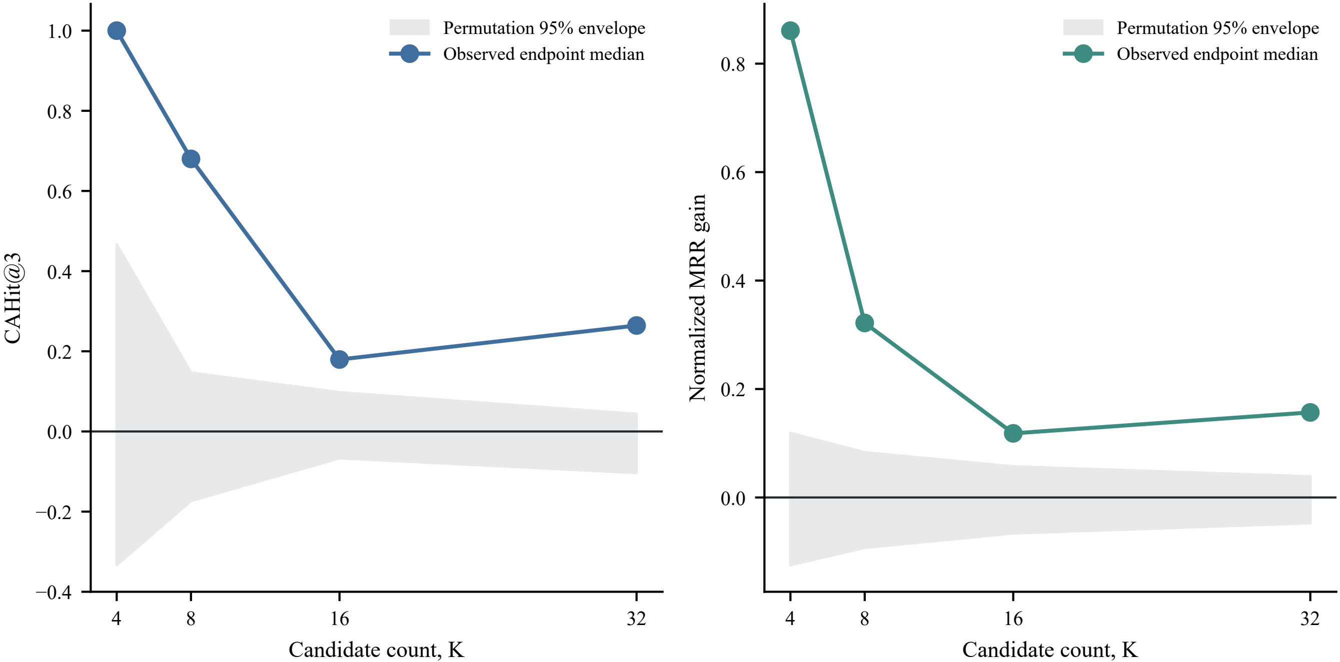


Figure S6. Graded signal-recovery positive control

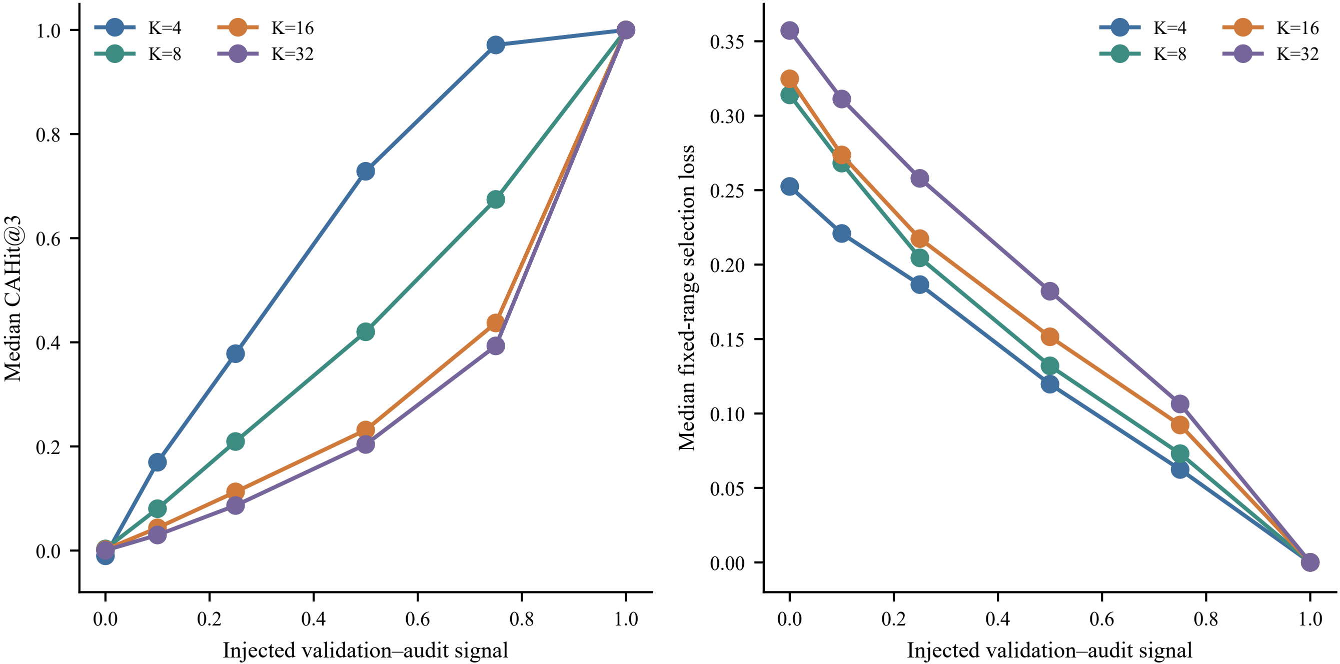


Figure S7. Full winner-optimism grid

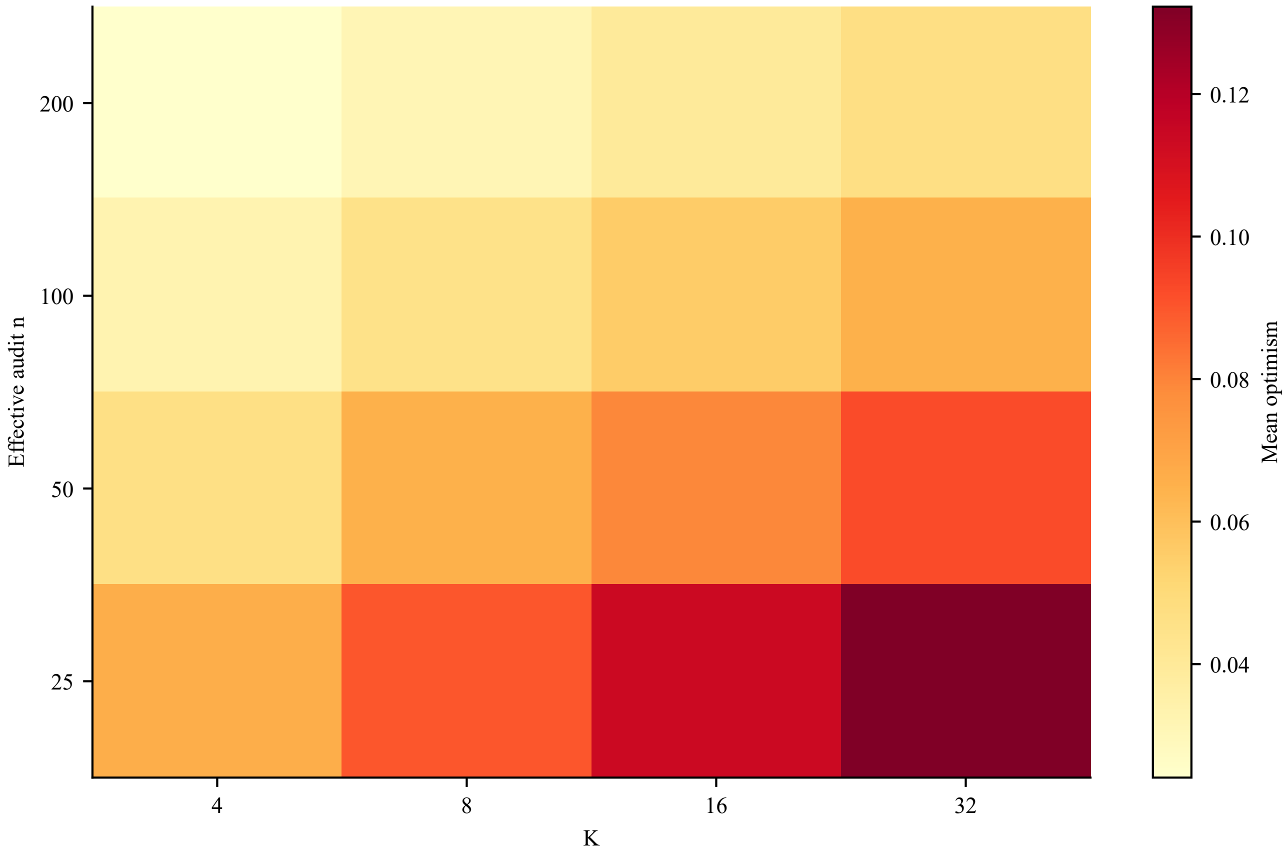


Figure S8. Fold-level audit gaps

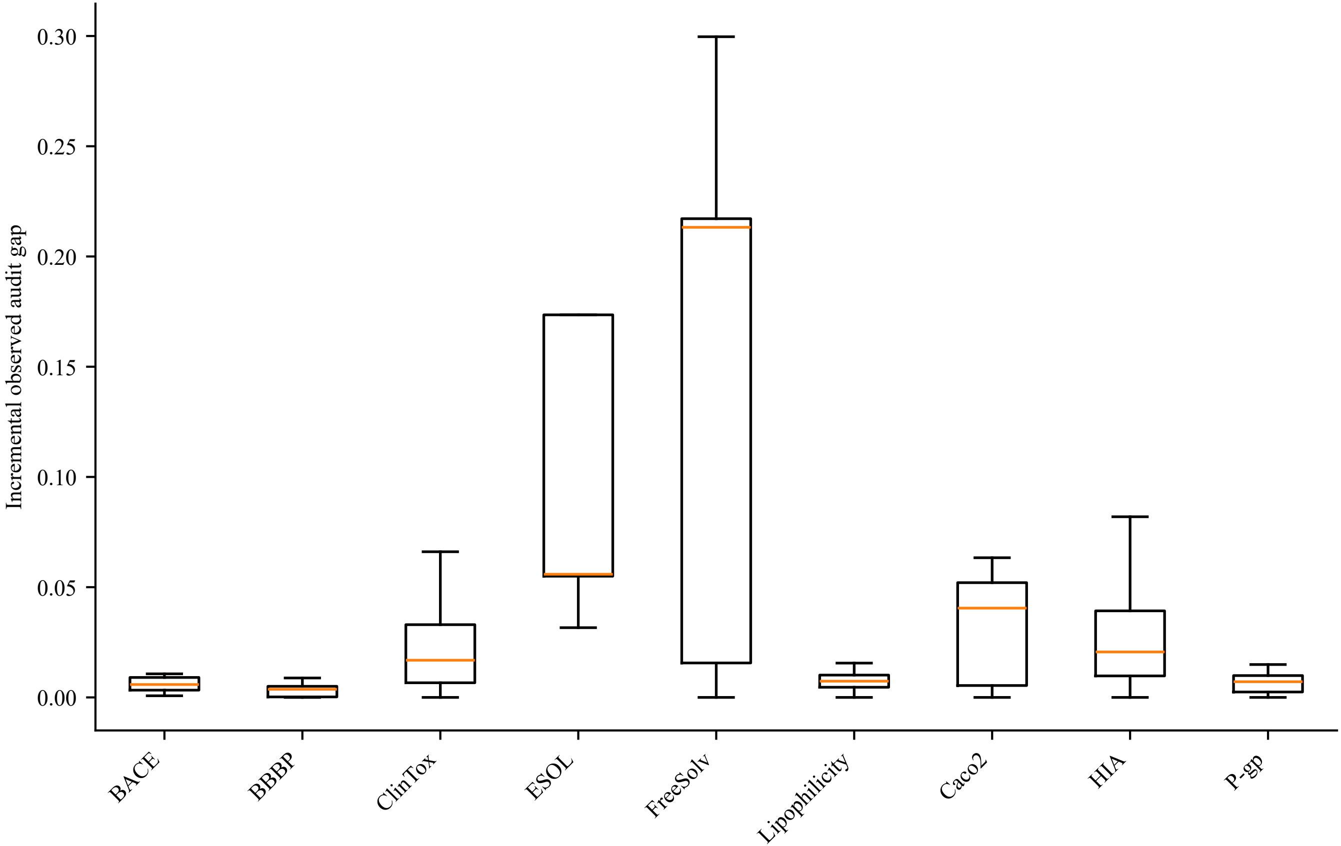


Figure S9. Full matched-K subset distributions

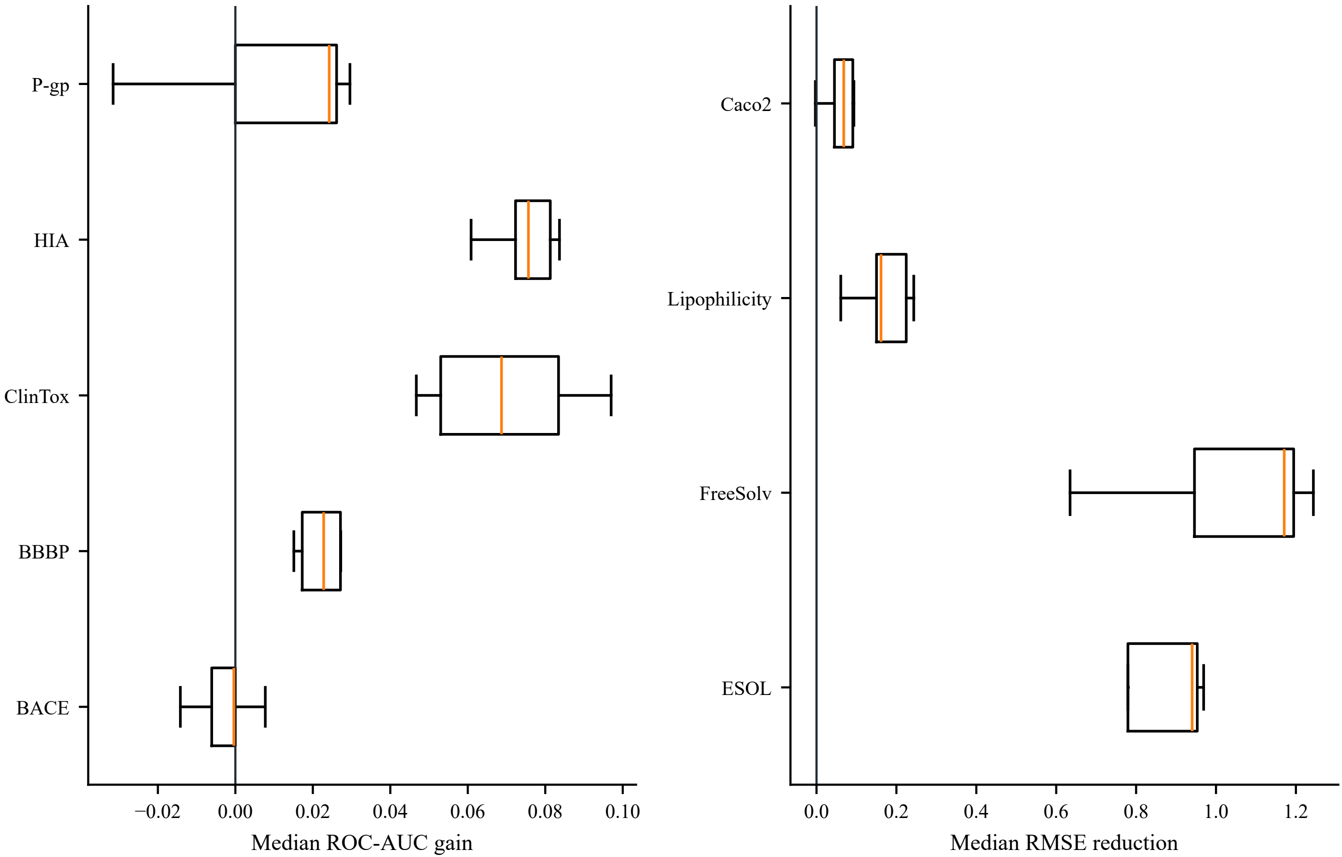


Figure S10. Representation and learner selection frequency

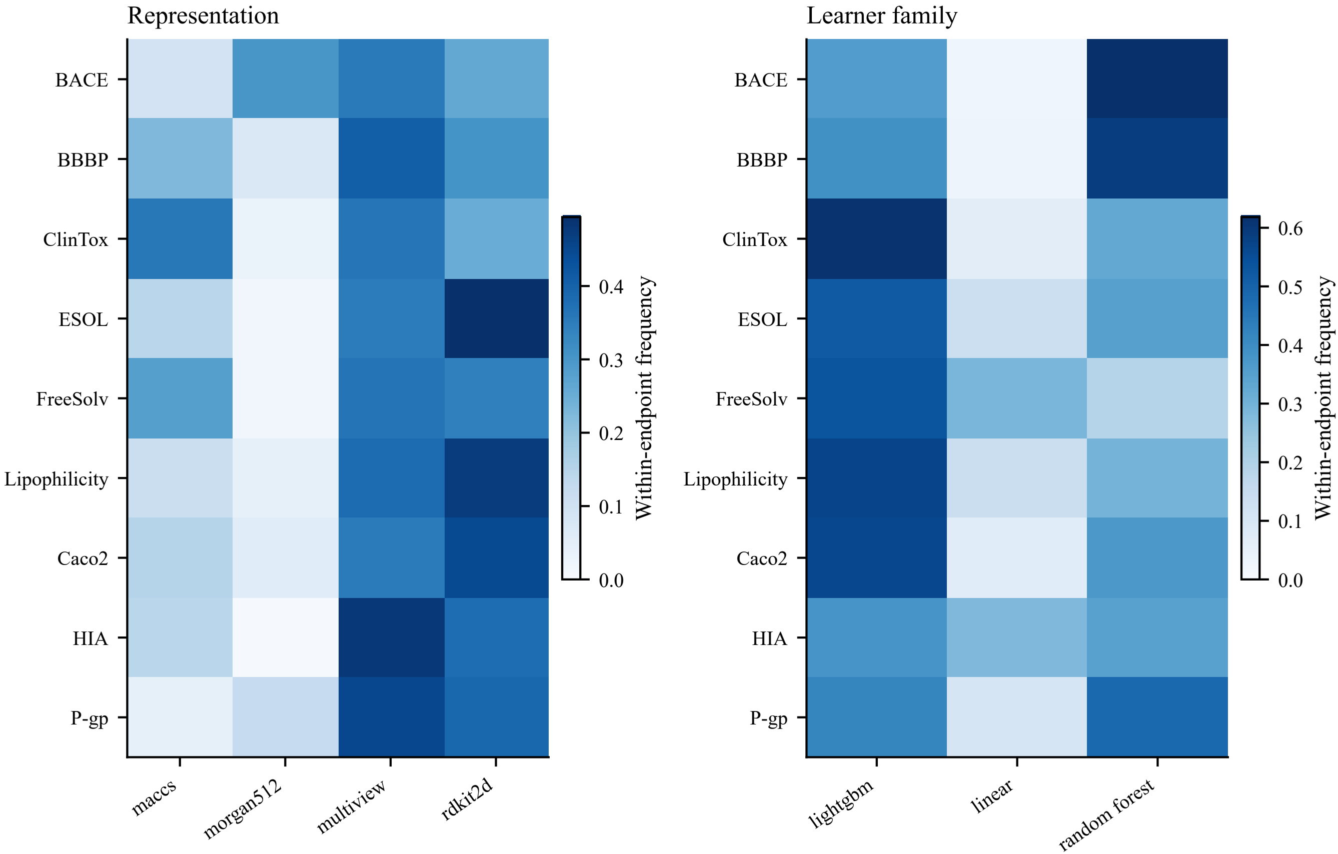


Figure S11. TDC endpoint panel

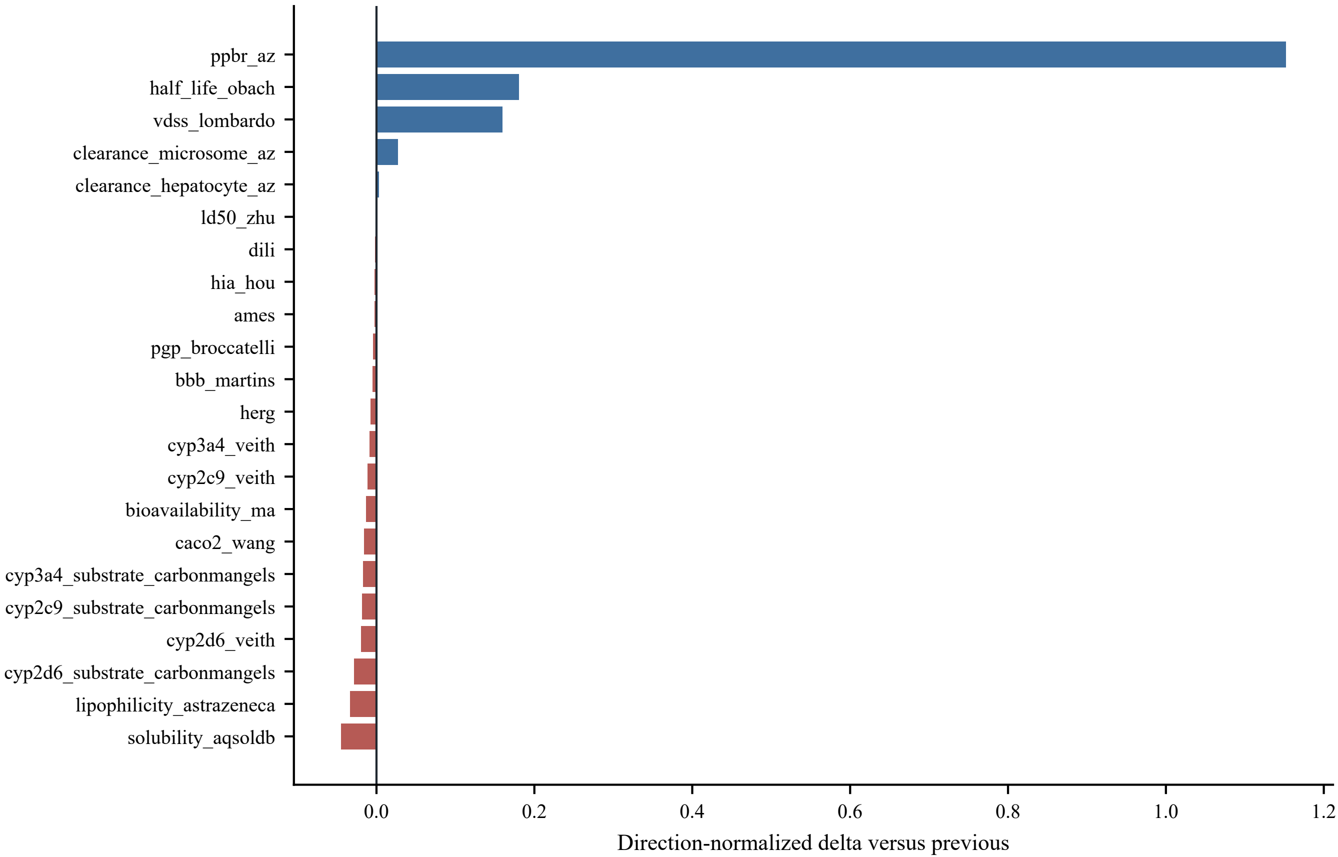


Figure S12. Conformal coverage and risk-ranking summaries

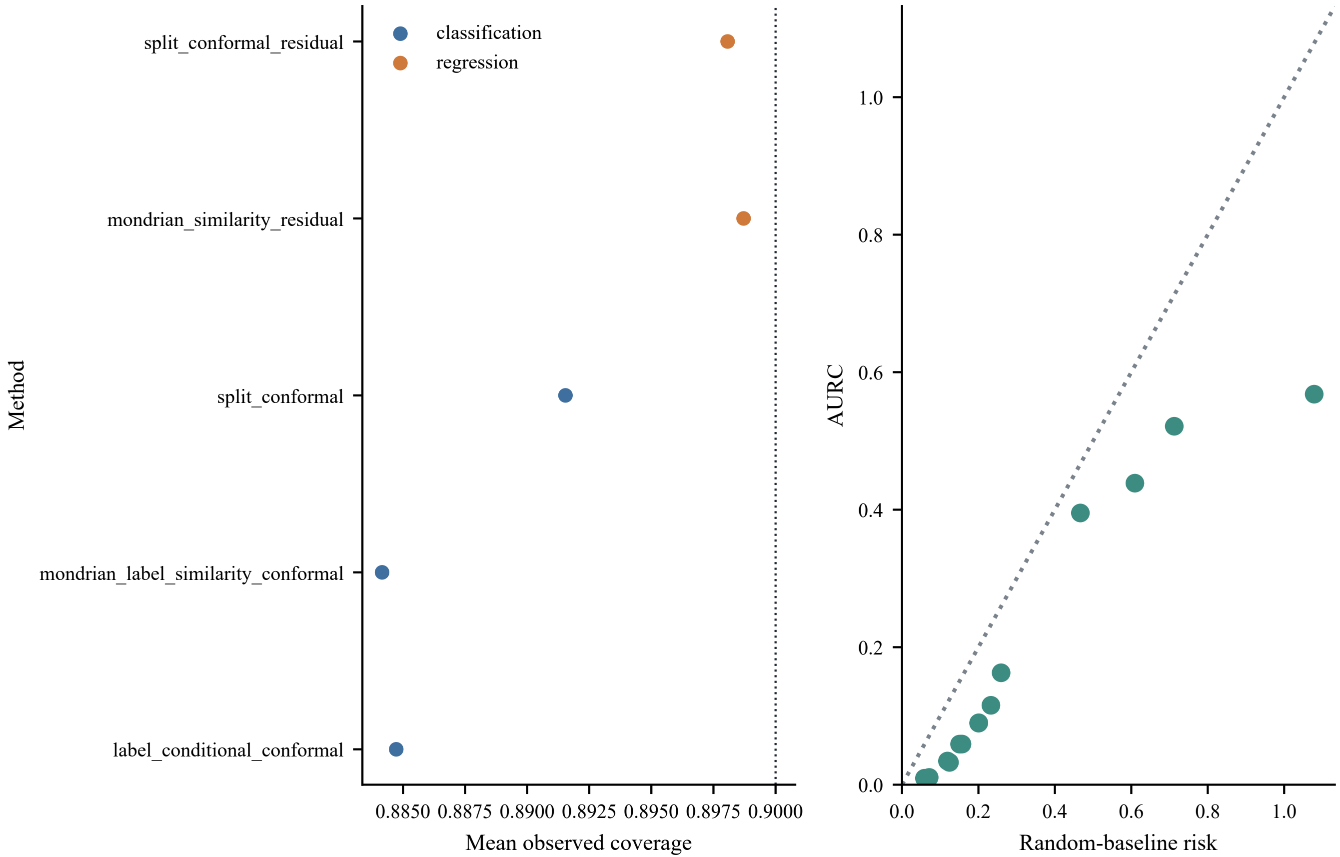


Figure S13. MoleculeACE activity-cliff results

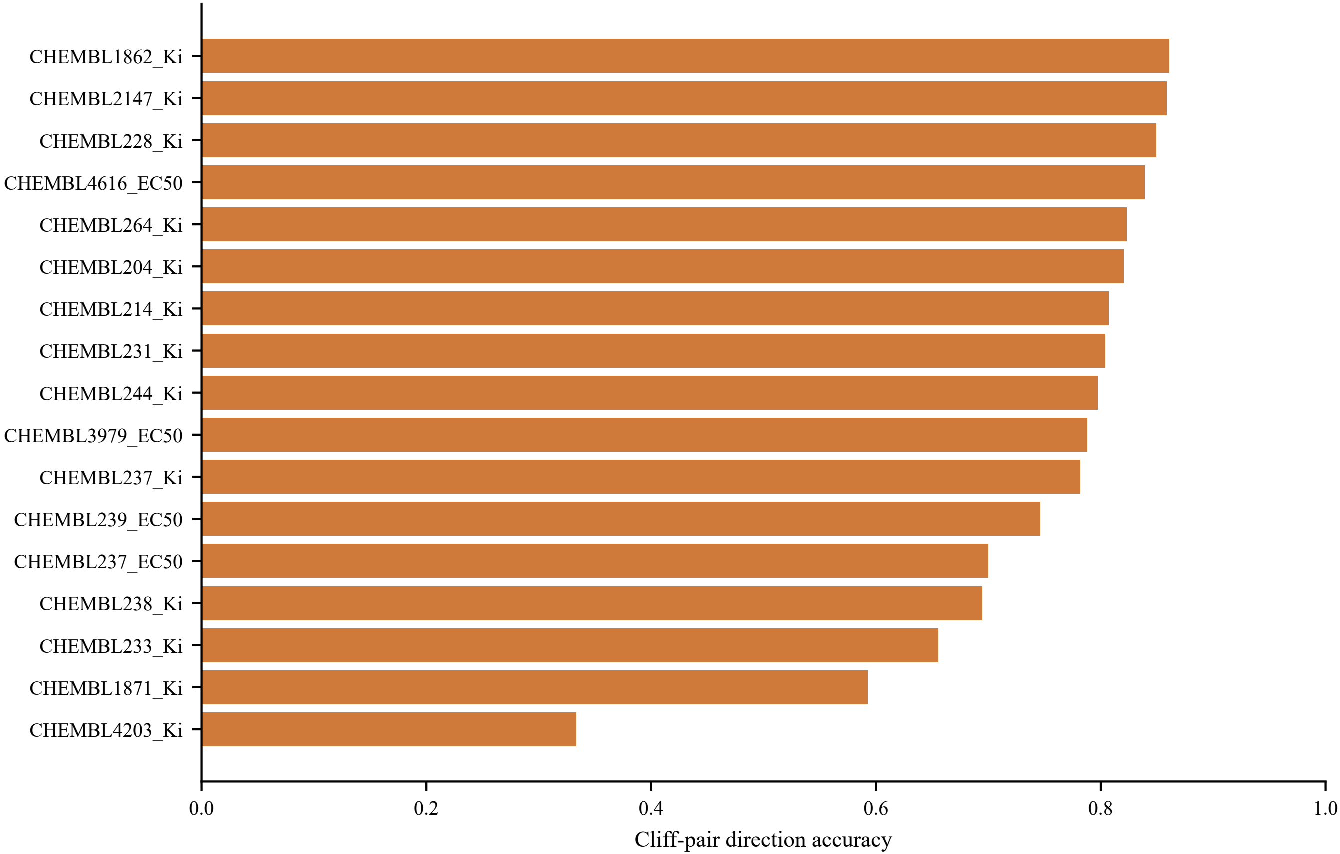


Figure S14. Beyond-rule-of-five migration results

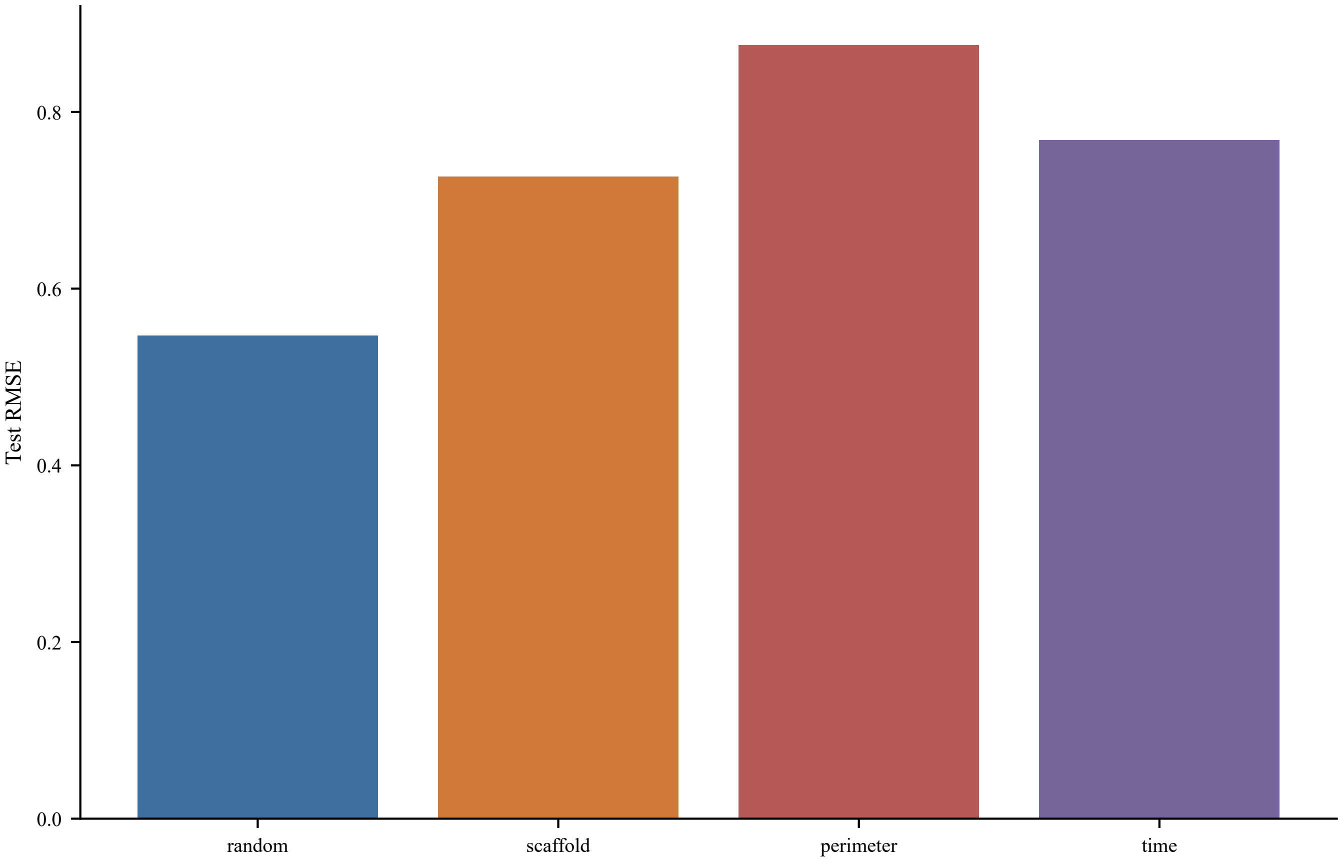
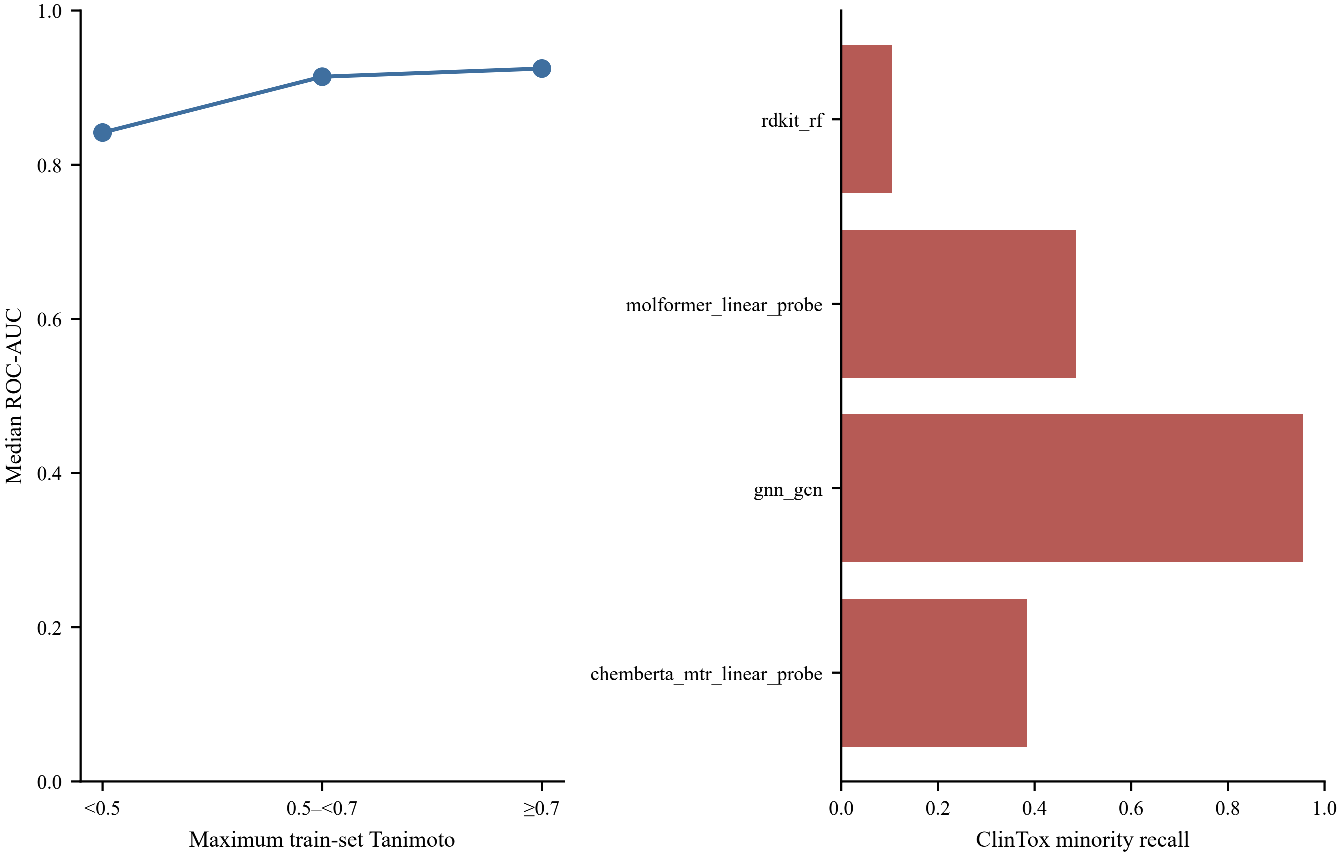
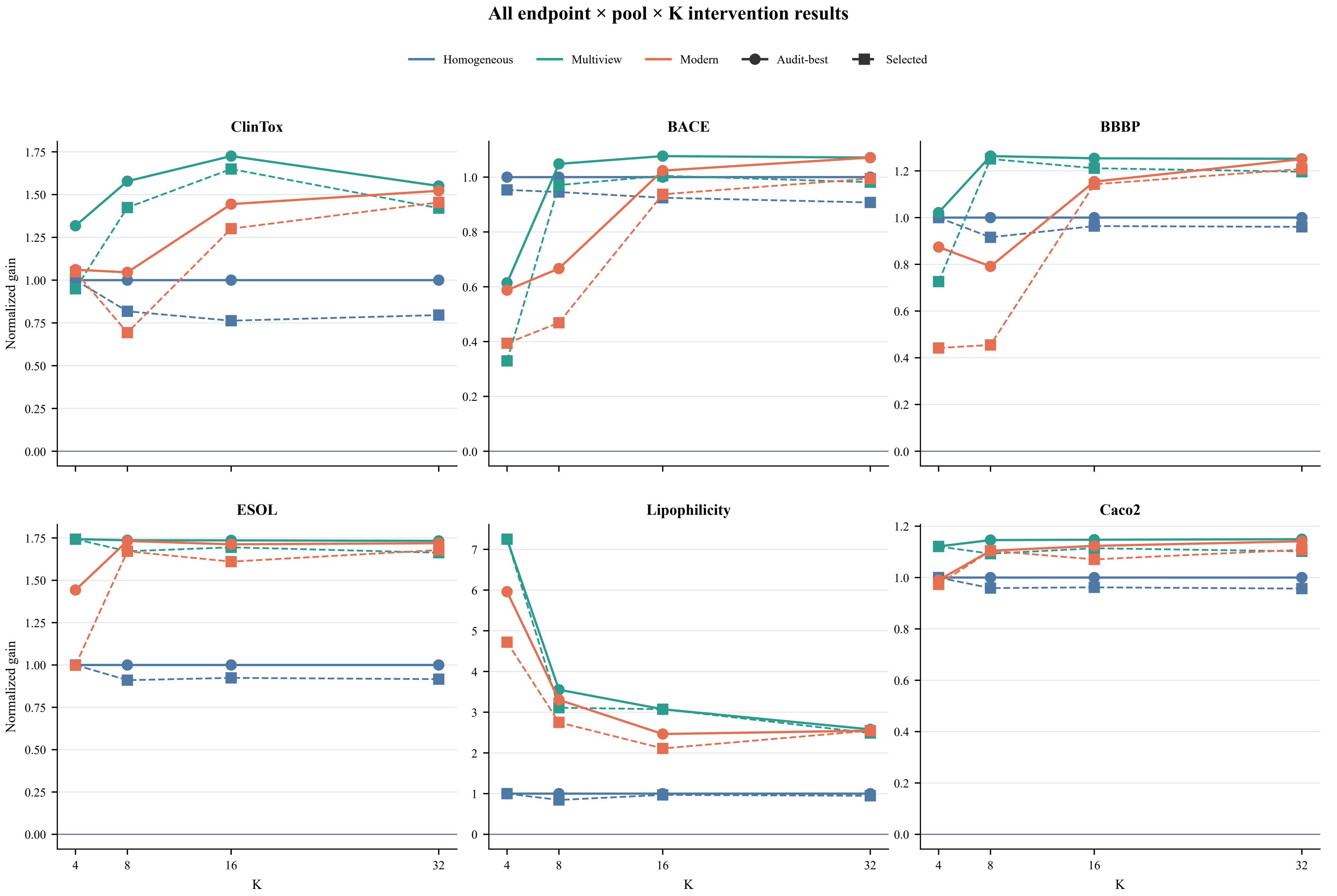


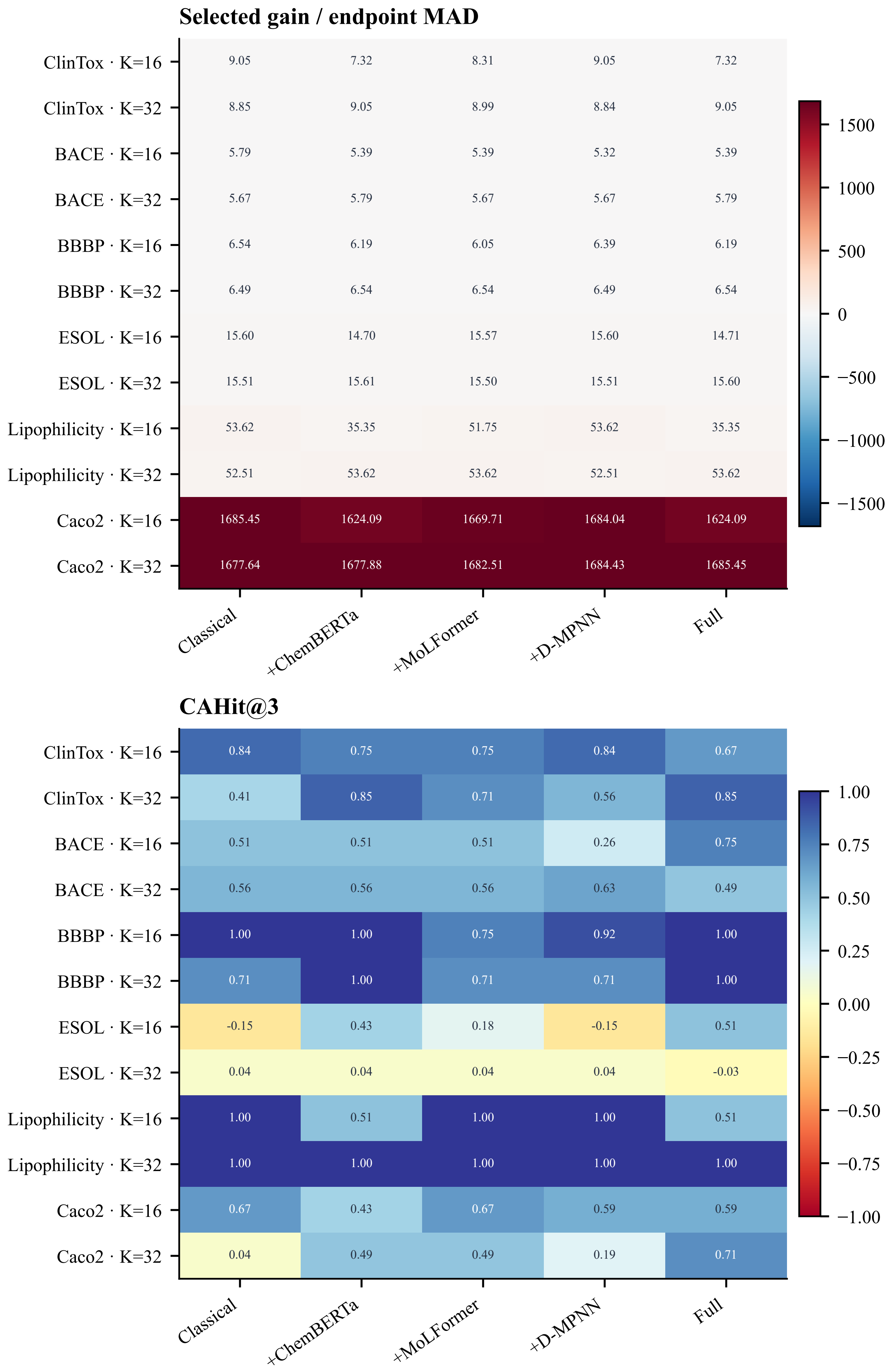
Figure S15. Low-support and ClinTox failure cases



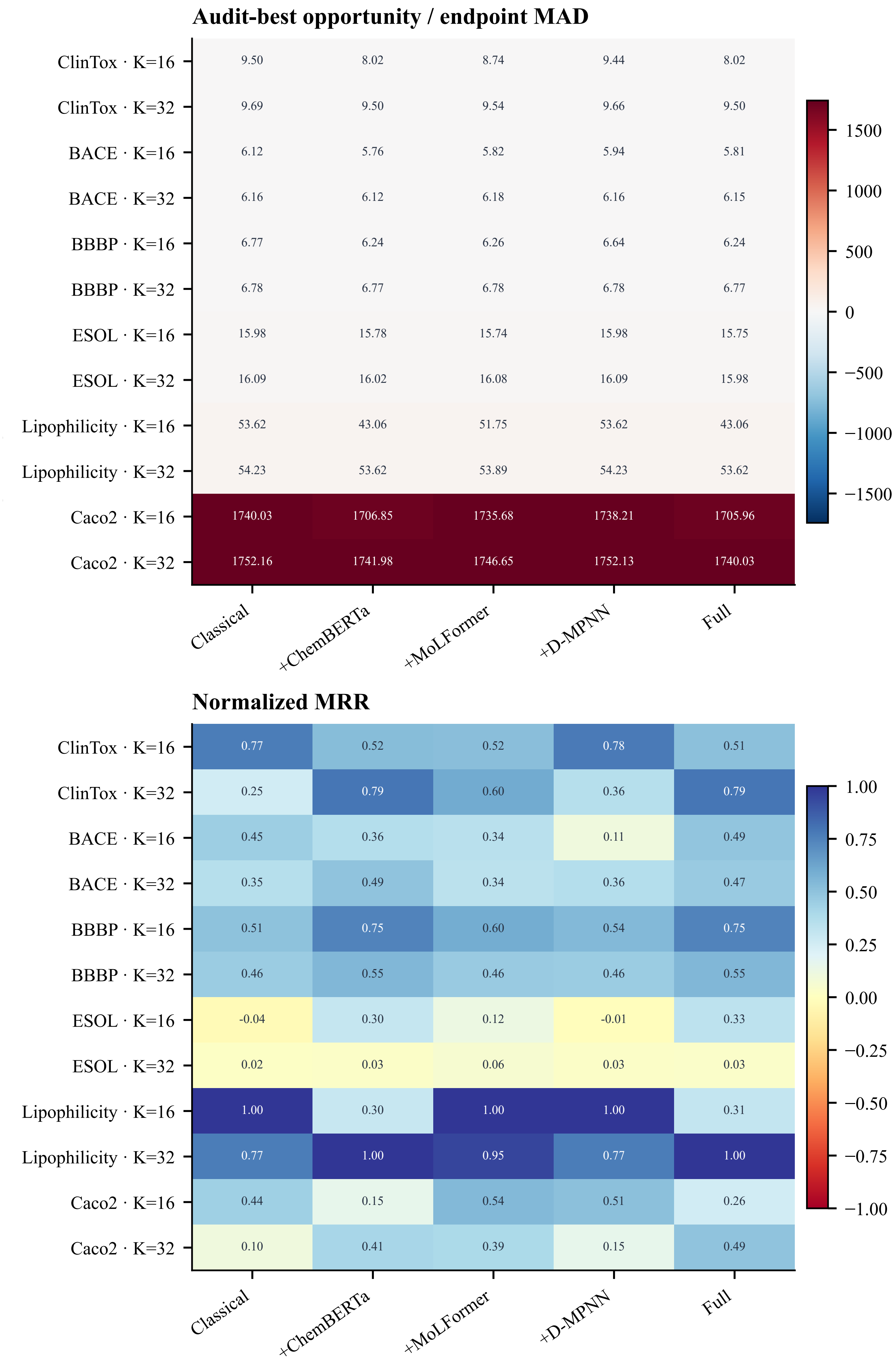
Supplementary Figure S16.



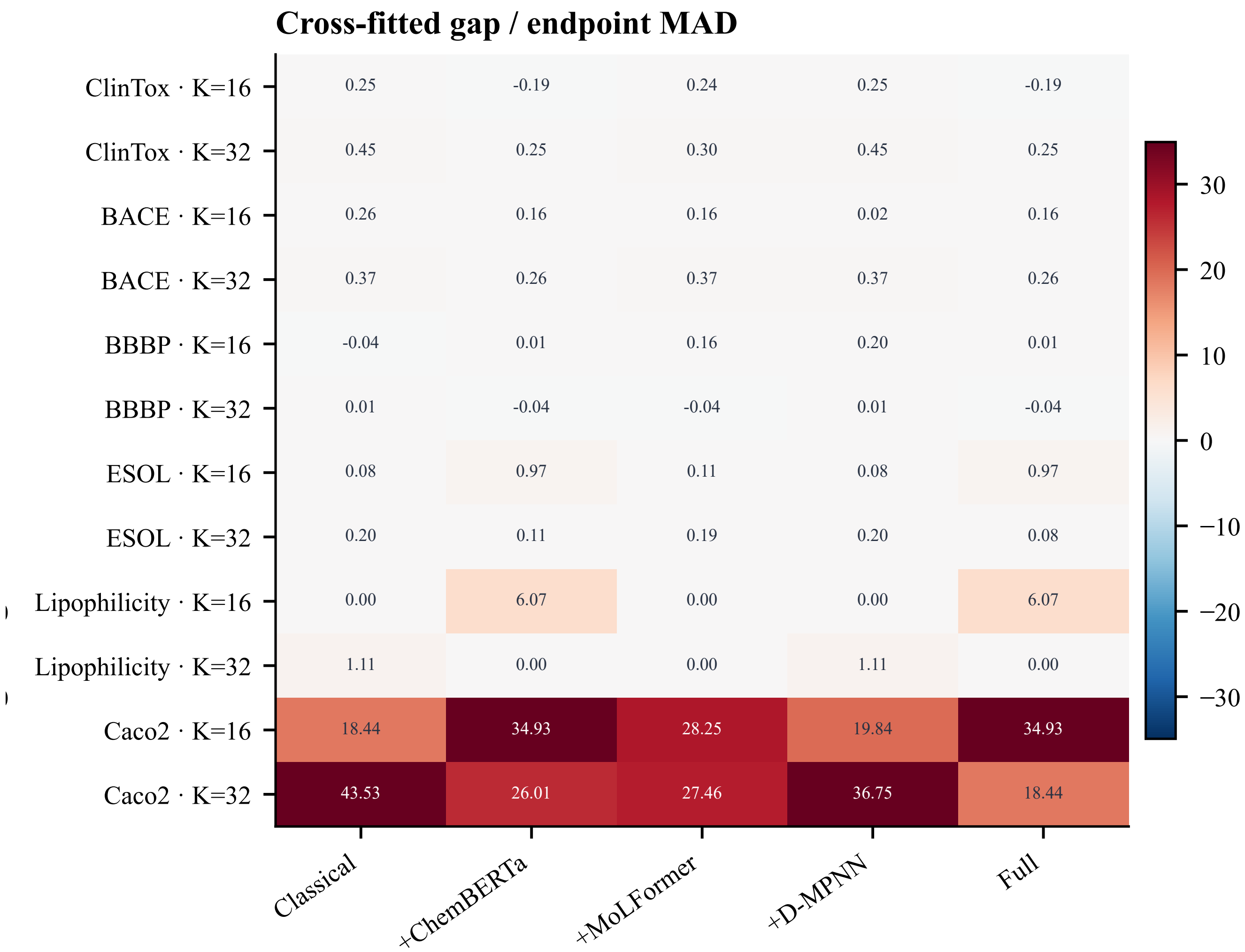
Supplementary Figure S17 (part 1 of 3): modern-component ablations

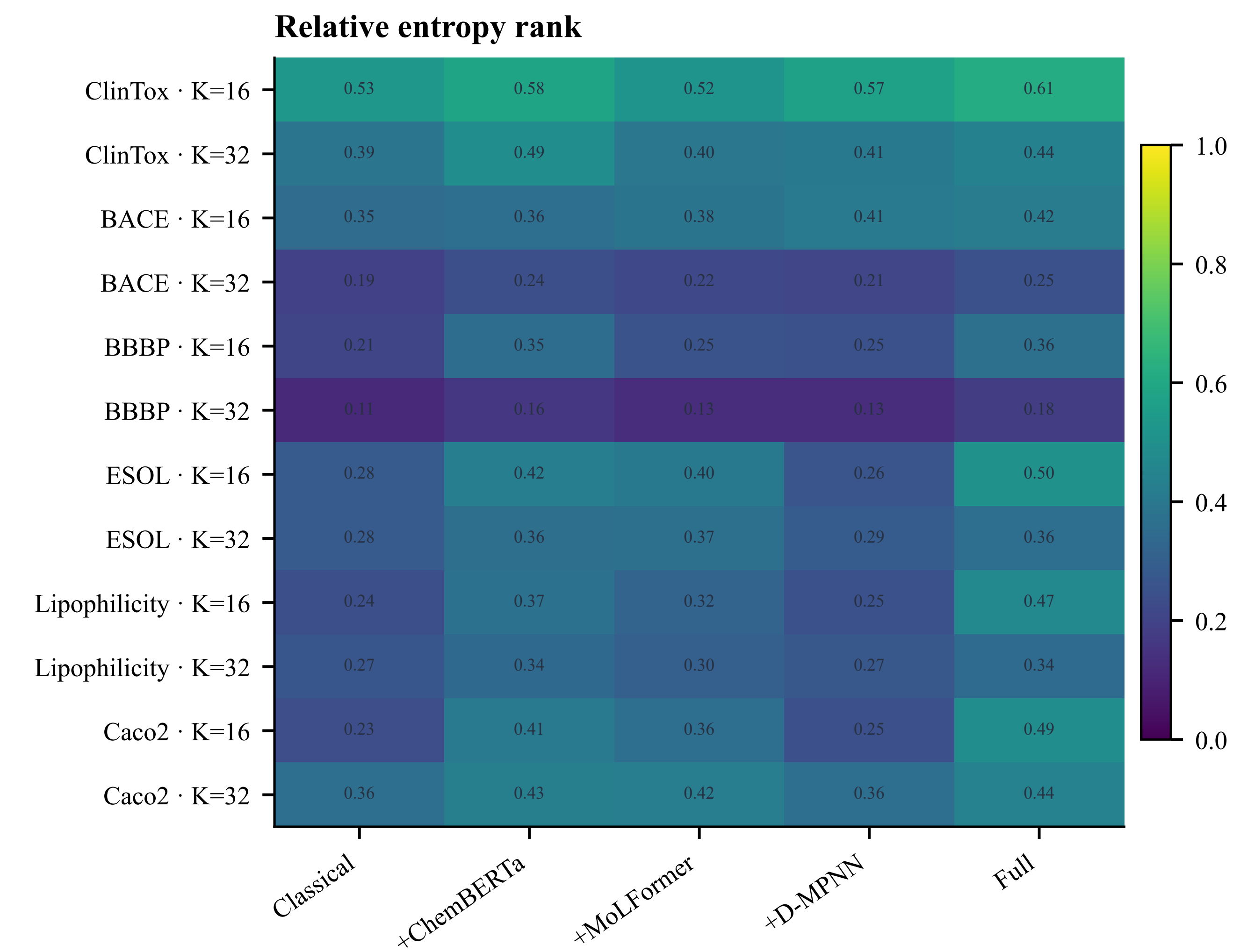


Supplementary Figure S17 (part 2 of 3): modern-component ablations

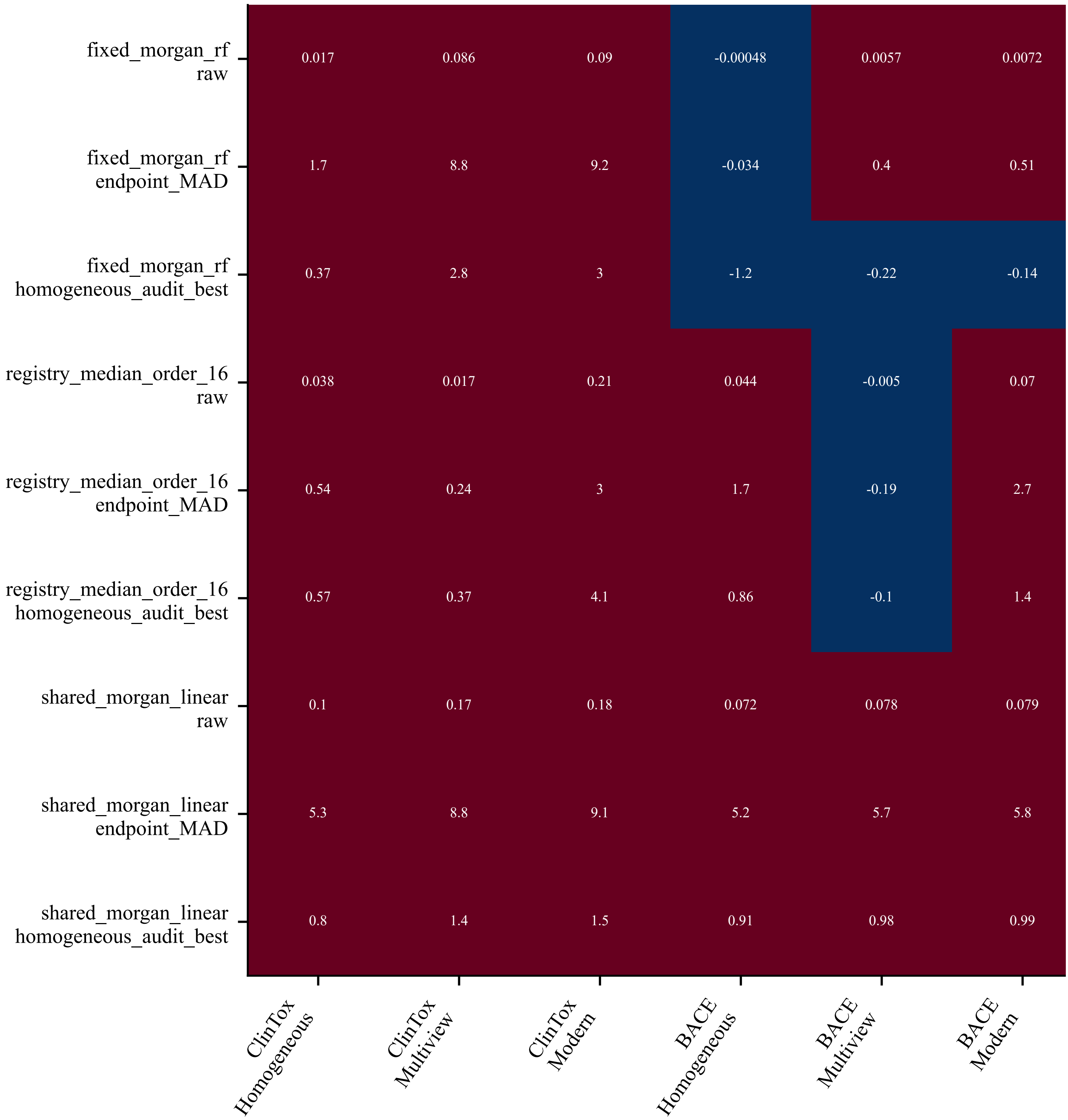


Supplementary Figure S17 (part 3 of 3): modern-component ablations

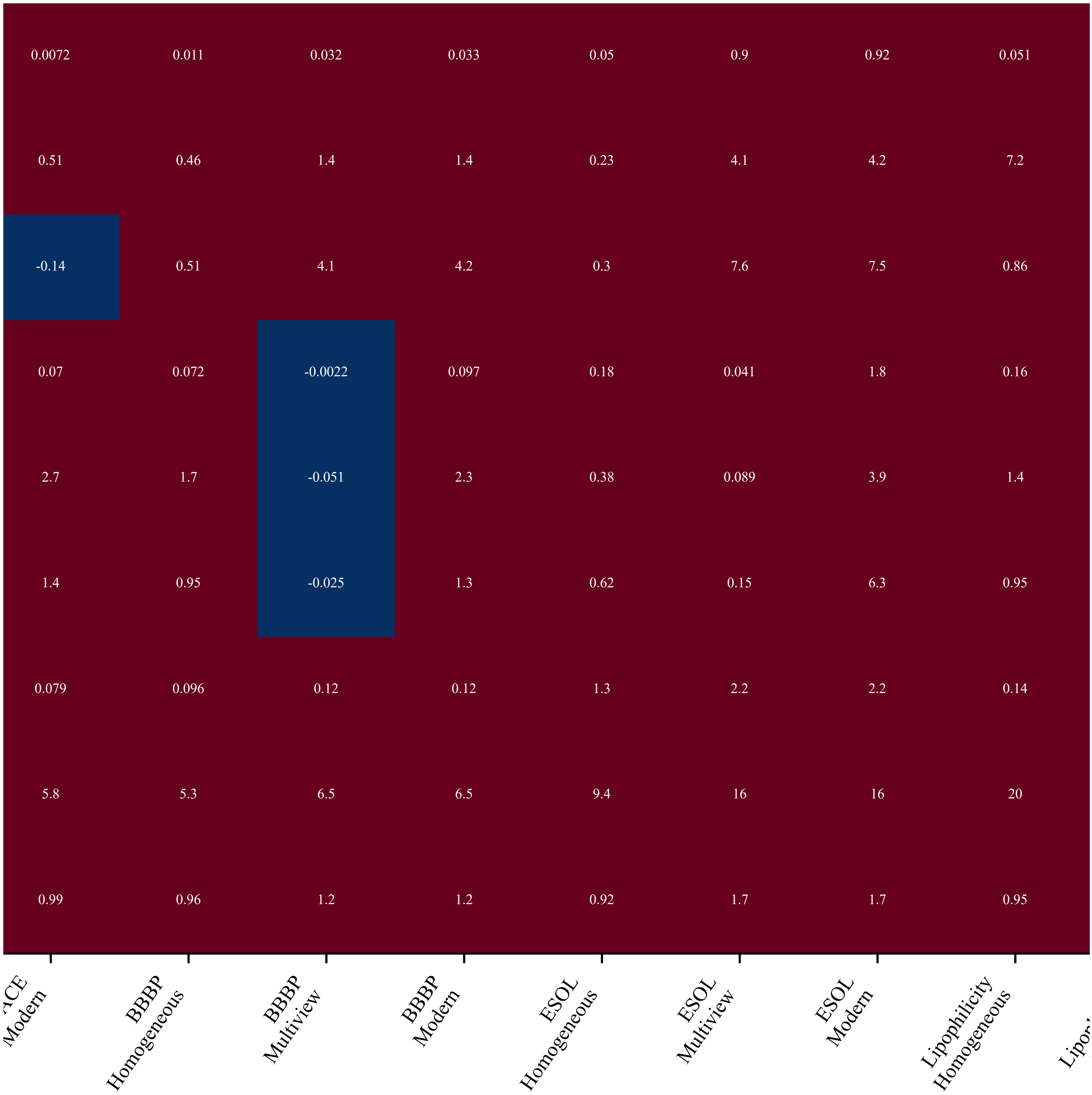




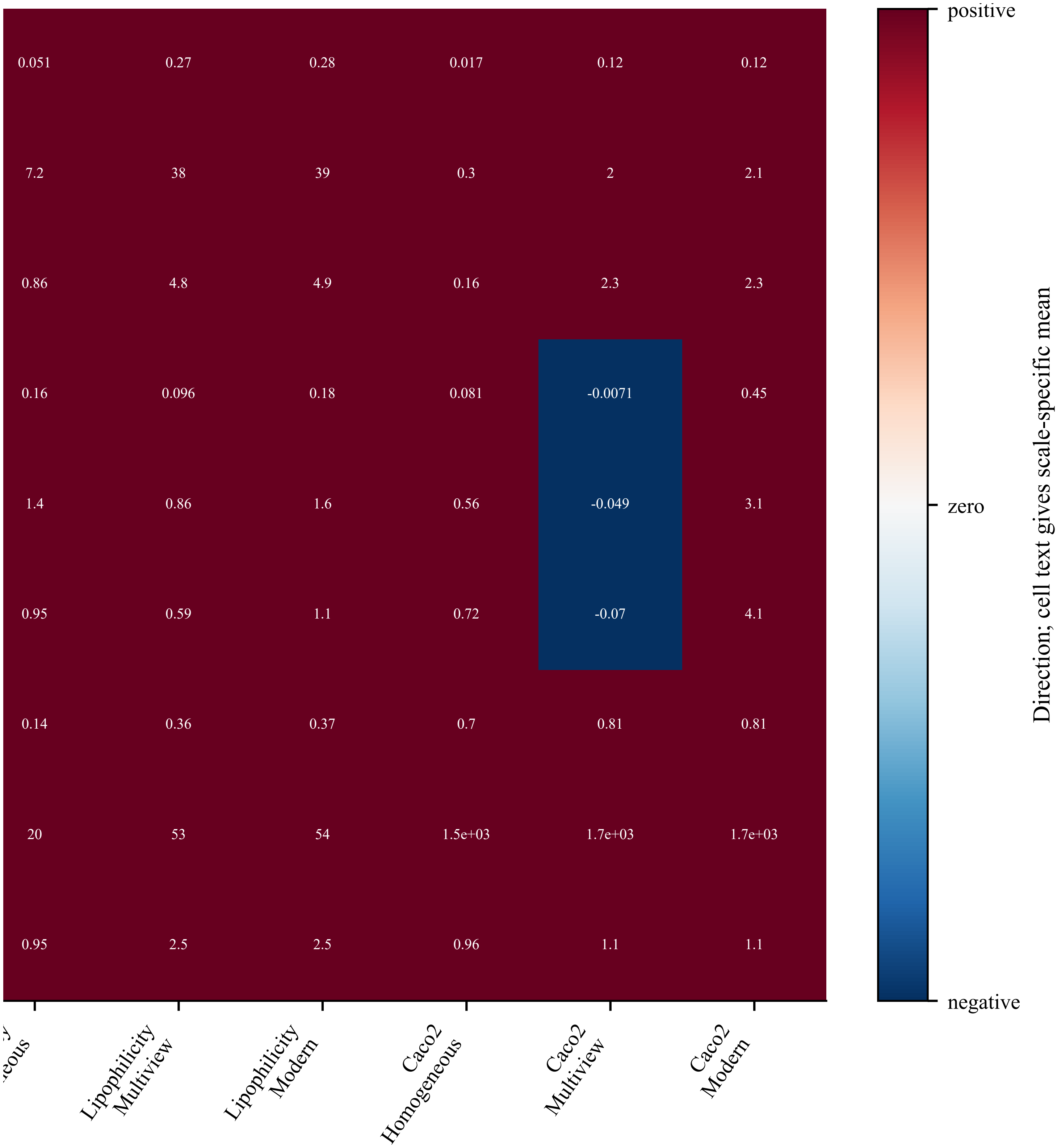
Supplementary Figure S18 (part 1 of 3): anchor and normalization sensitivity



Supplementary Figure S18 (part 2 of 3): anchor and normalization sensitivity

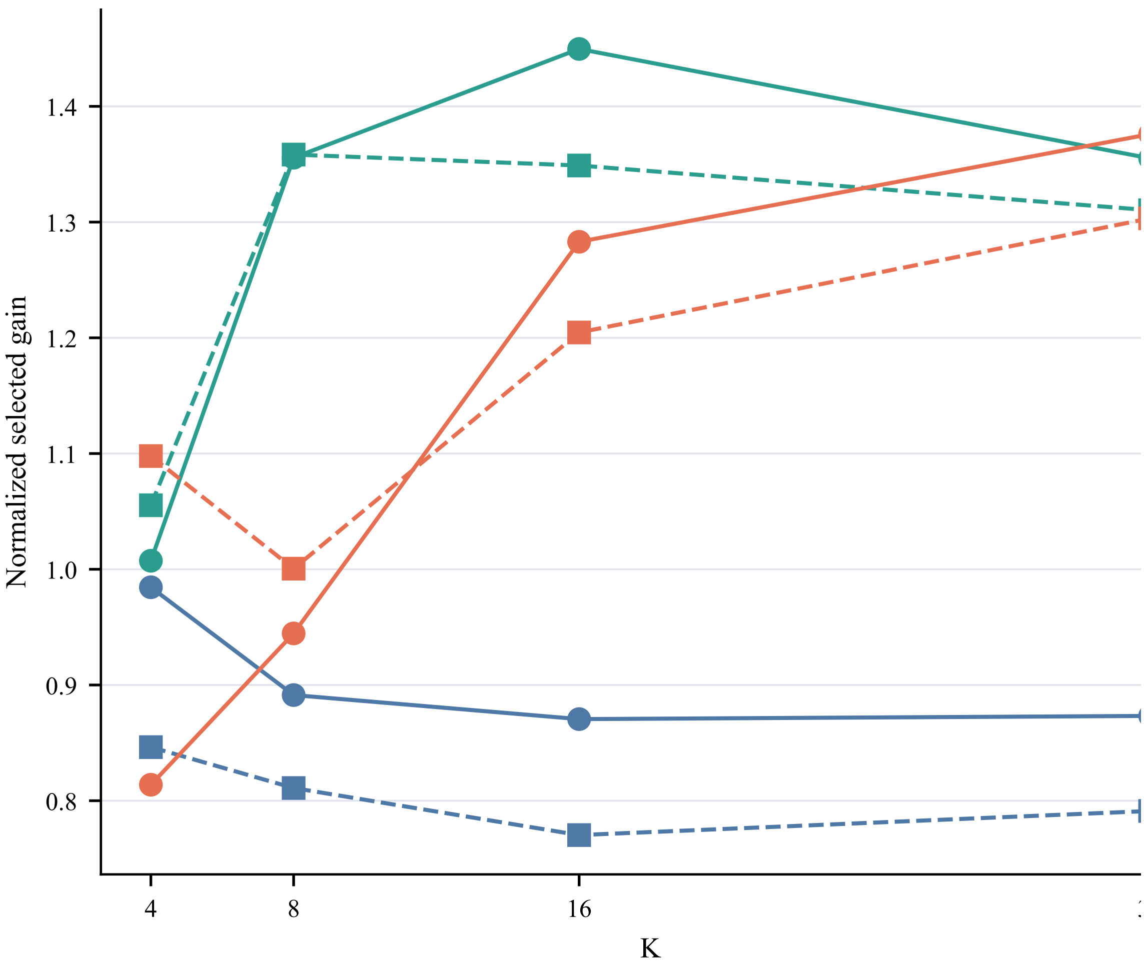


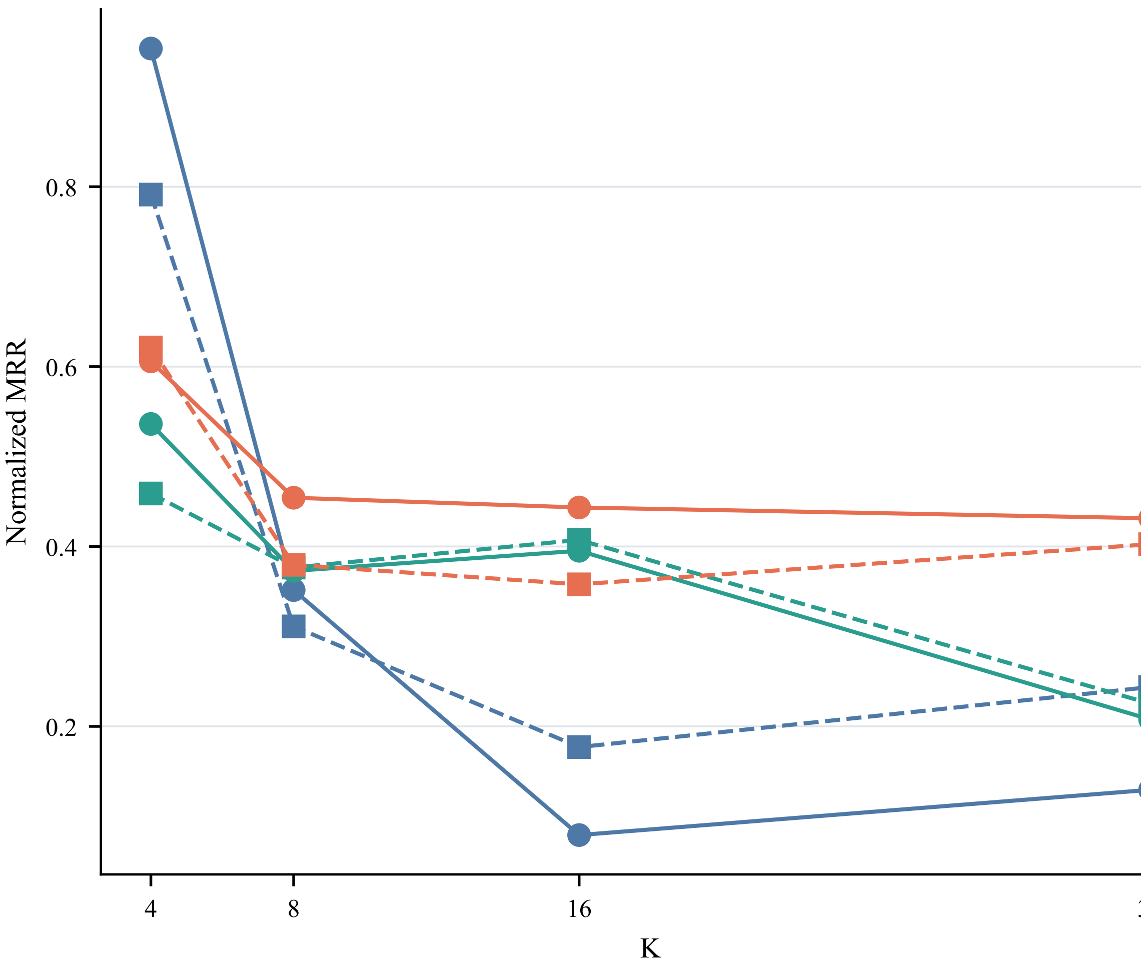
Supplementary Figure S18 (part 3 of 3): anchor and normalization sensitivity



Supplementary Figure S19 (part 1 of 3): split-regime transport

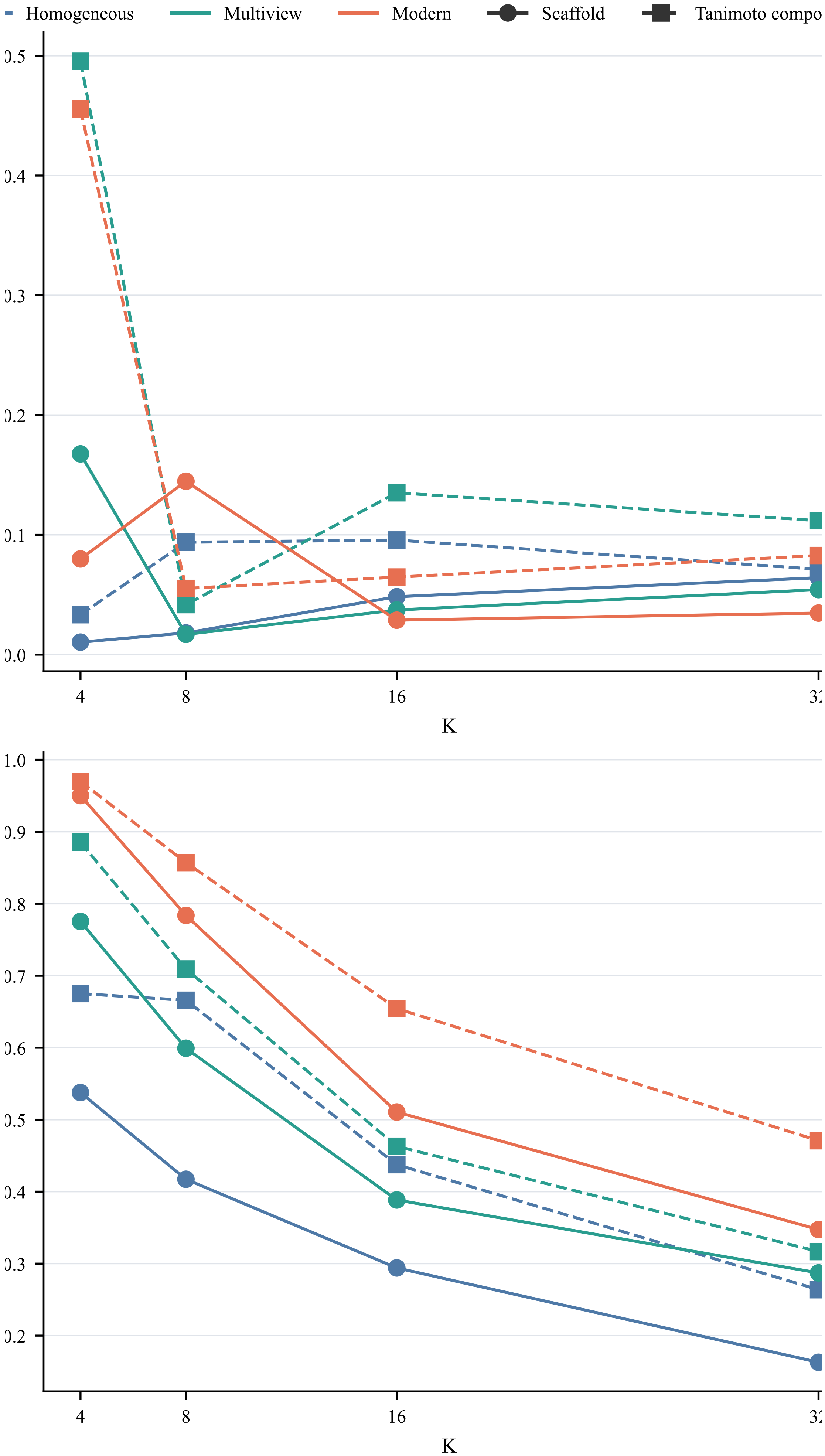
Colors: homogeneous, multiview, modern; solid circles: scaffold; dashed squares: Tanimoto component.





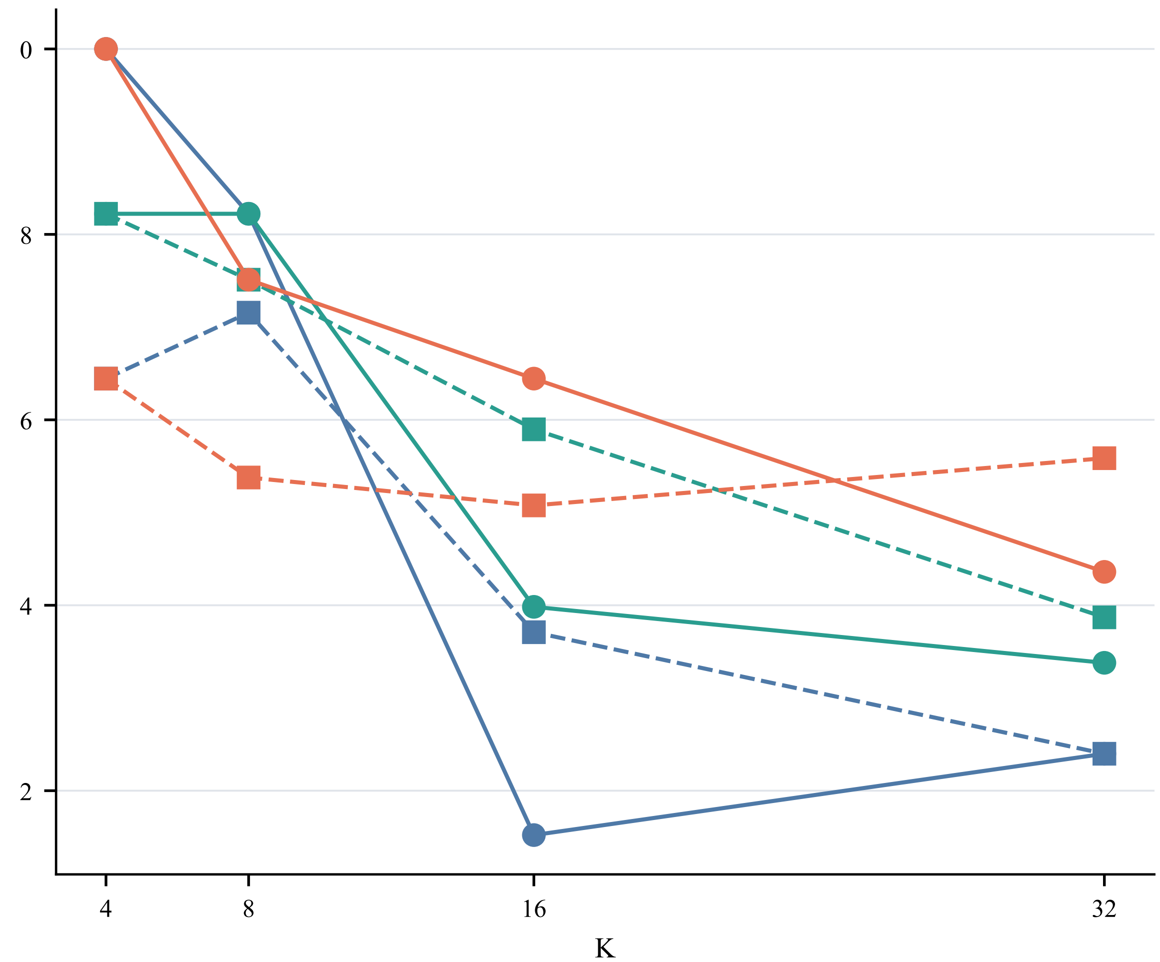
Supplementary Figure S19 (part 2 of 3): split-regime transport

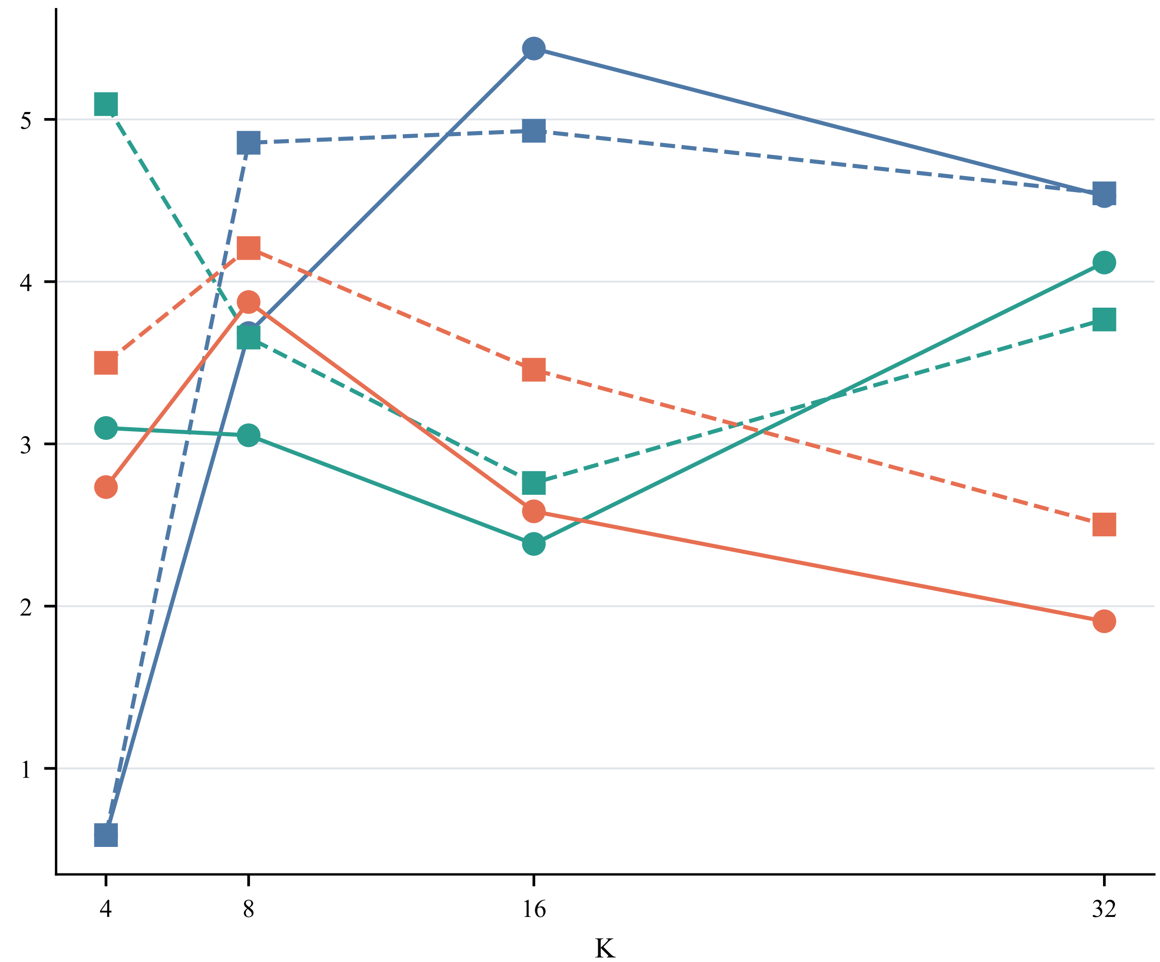
Colors: homogeneous, multiview, modern; solid circles: scaffold; dashed squares: Tanimoto component.



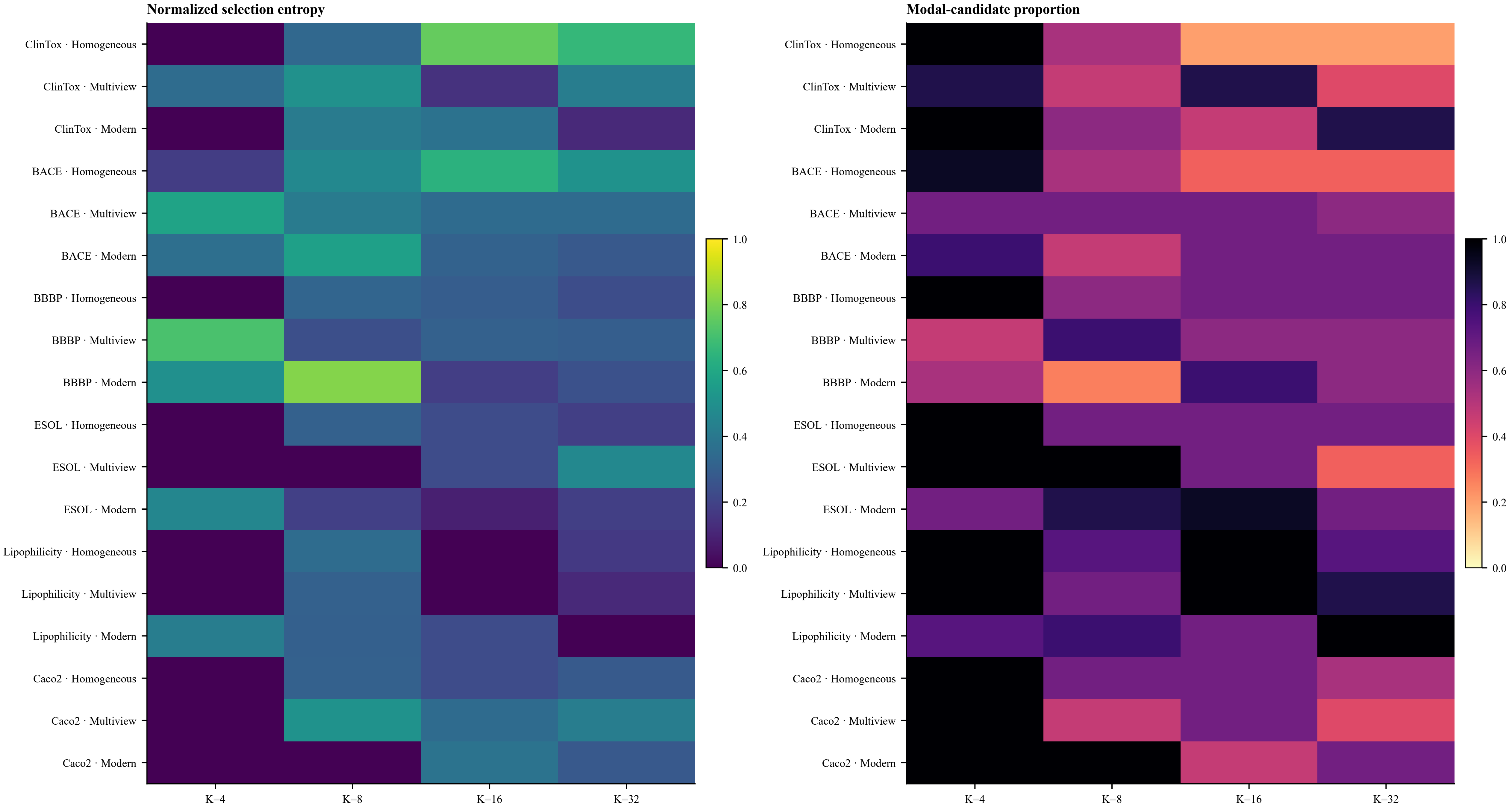
Supplementary Figure S19 (part 3 of 3): split-regime transport

Colors: homogeneous, multiview, modern; solid circles: scaffold; dashed squares: Tanimoto component.

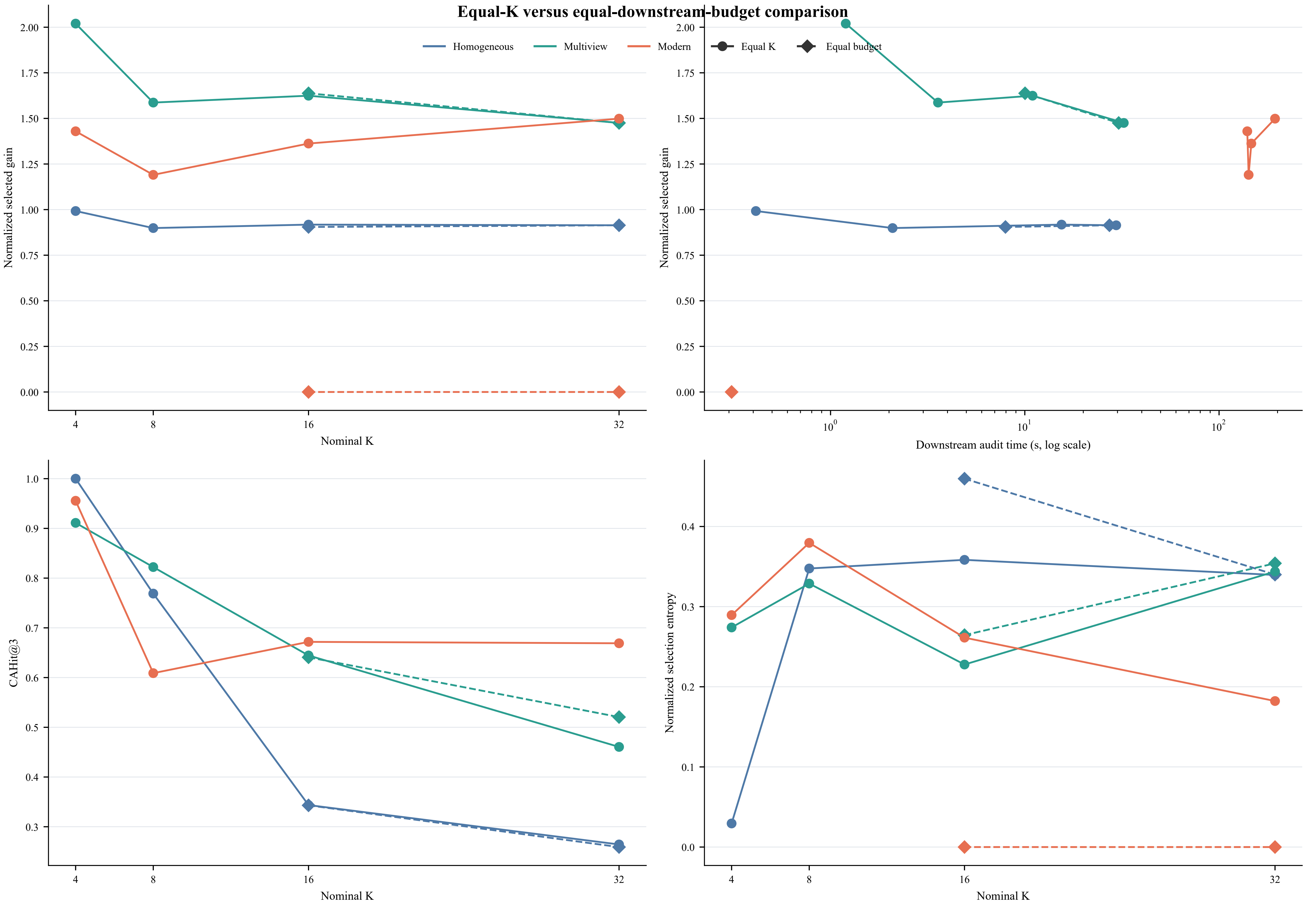




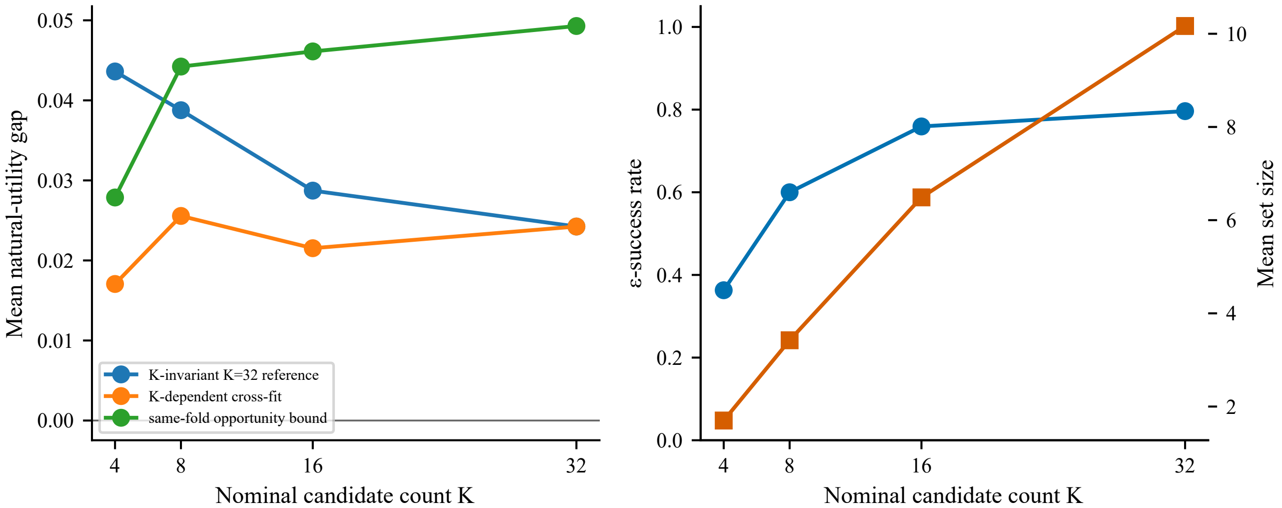
Supplementary Figure S20.



Supplementary Figure S21.



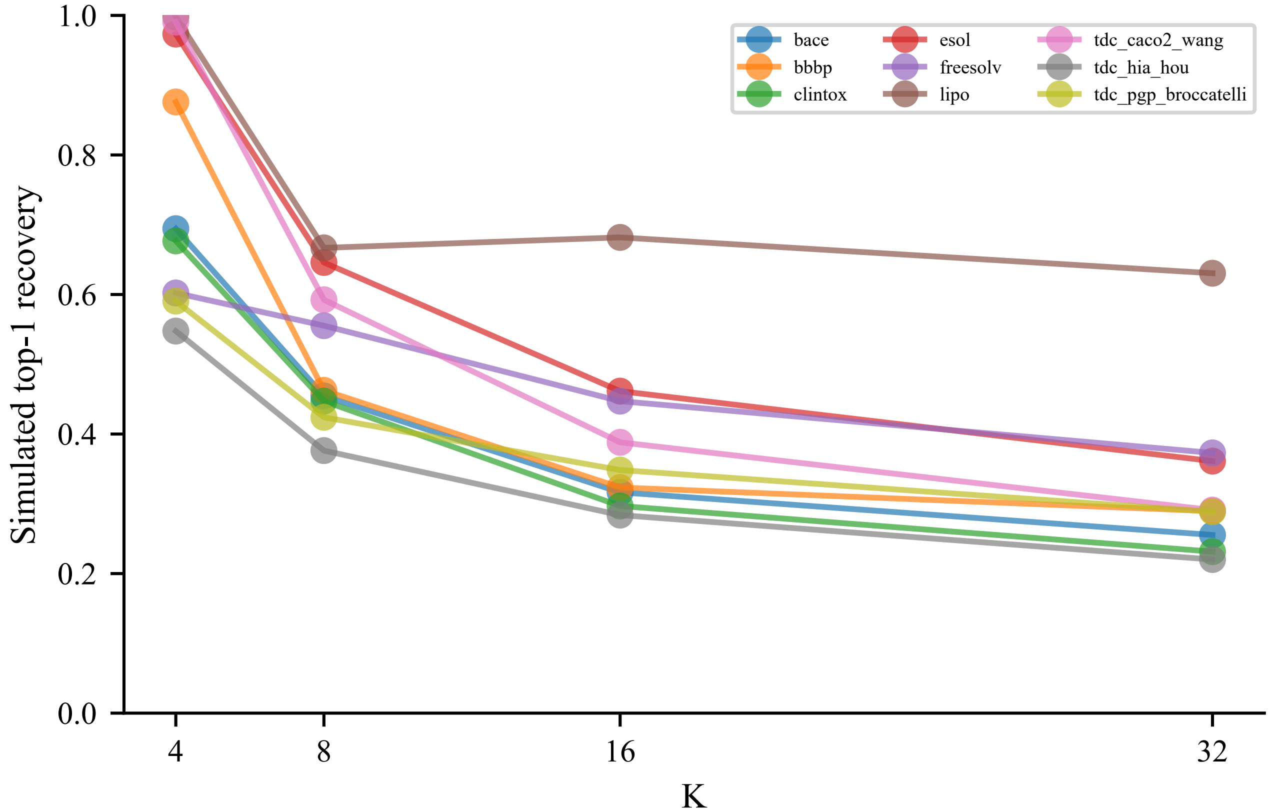
Supplementary Figure S22. Practical equivalence and metric-dependent selection



K-invariant, K-dependent and same-fold gaps are contrasted with ε-success and near-equivalent-set summaries.

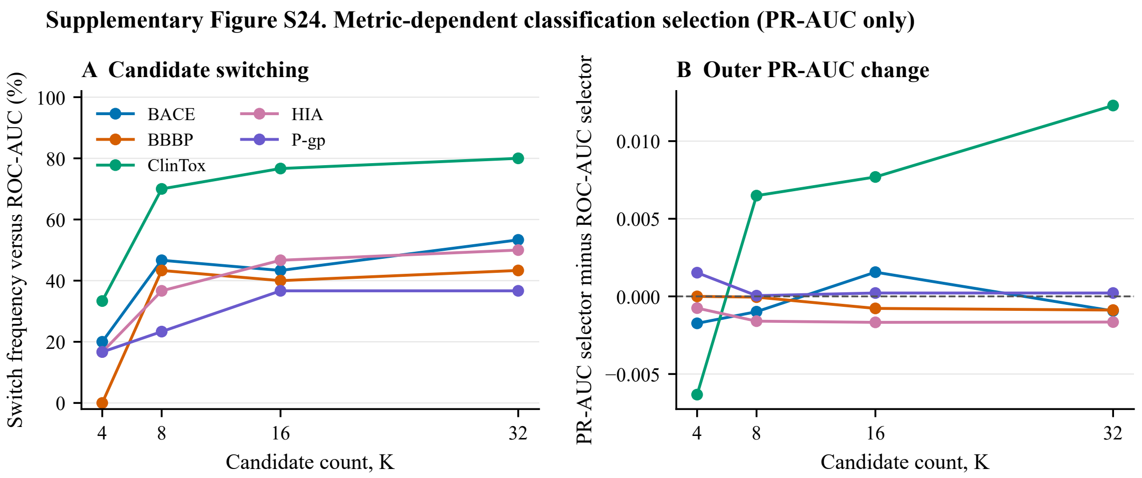
Supplementary Figure S23.

Supplementary Figure S23. Empirical correlated heteroskedastic recovery



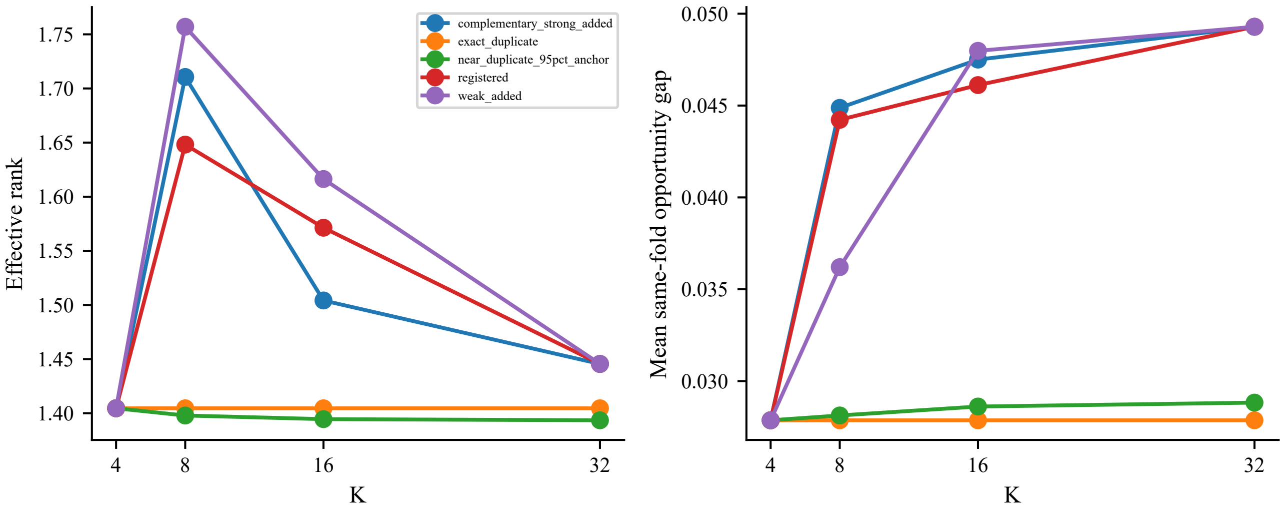
Top-1 recovery under endpoint-specific empirical residual covariance at the estimated noise scale; 2 000 replicates per endpoint and K.

Supplementary Figure S24.



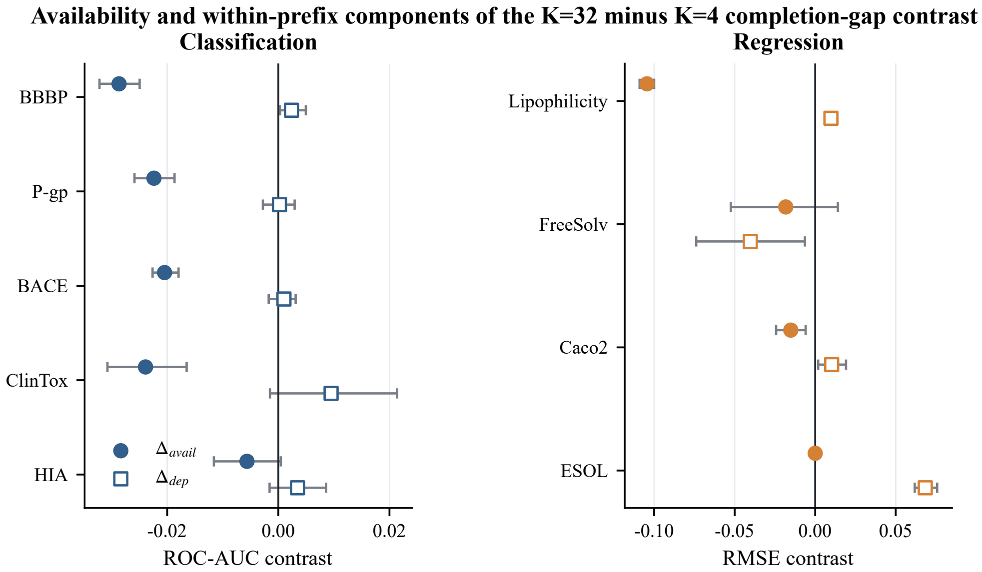
Recall-constrained effects are excluded because frozen-identity inner-validation probabilities were not retained.

Supplementary Figure S25. Constructed candidate-dependence controls

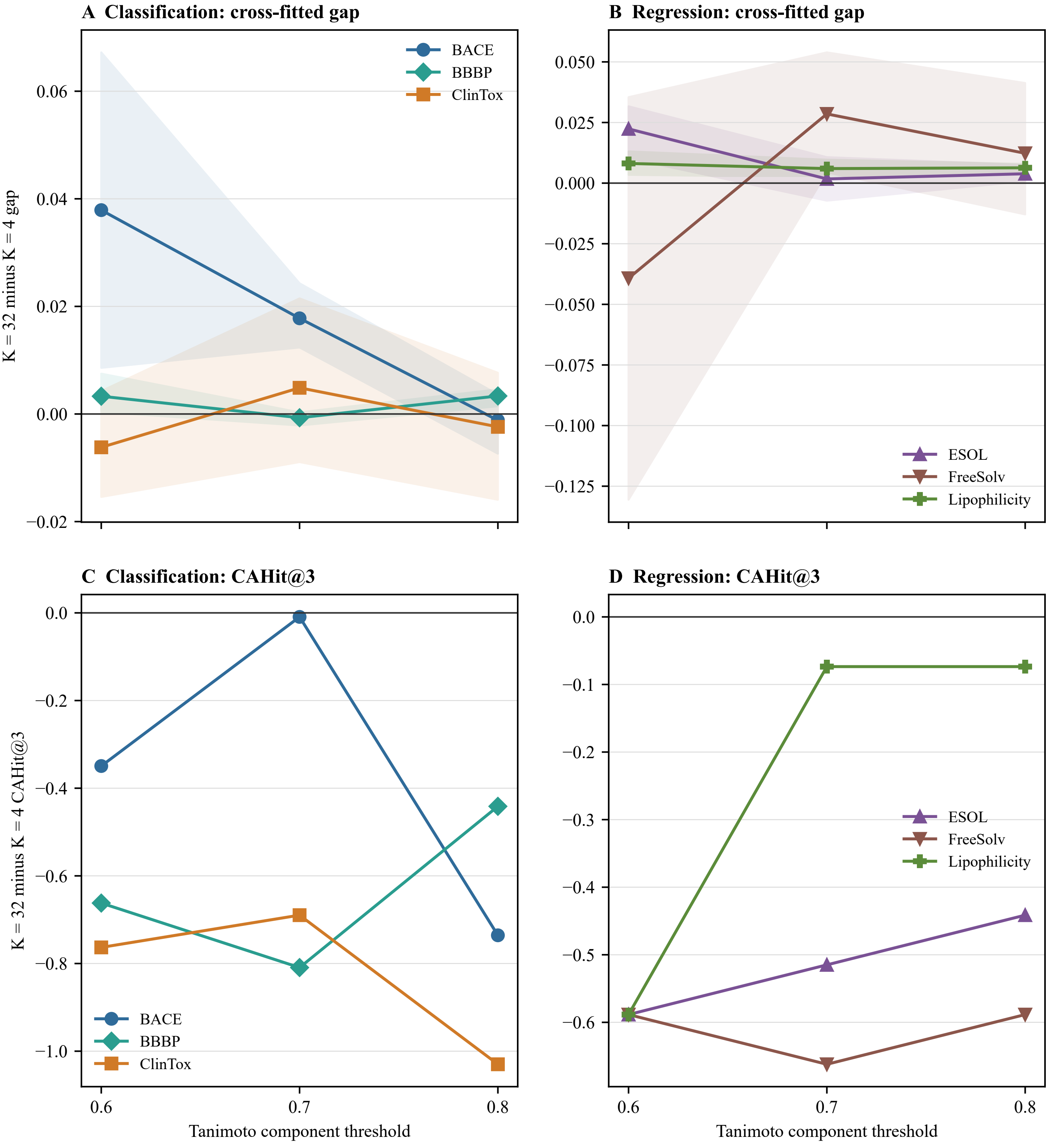


Exact duplicates, 95% anchor near-duplicates, weak additions, complementary-strong ordering and the registered pool distinguish nominal K from utility-pattern diversity.

Supplementary Figure S26.



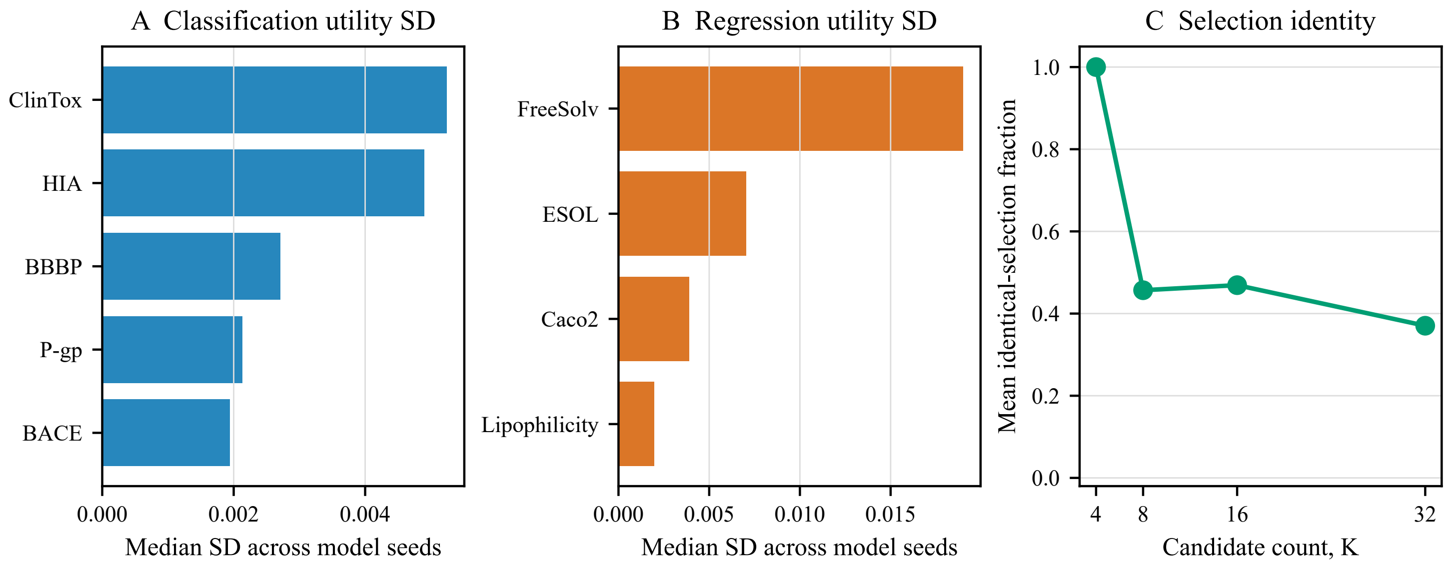
Supplementary Figure S27. Component-threshold sensitivity



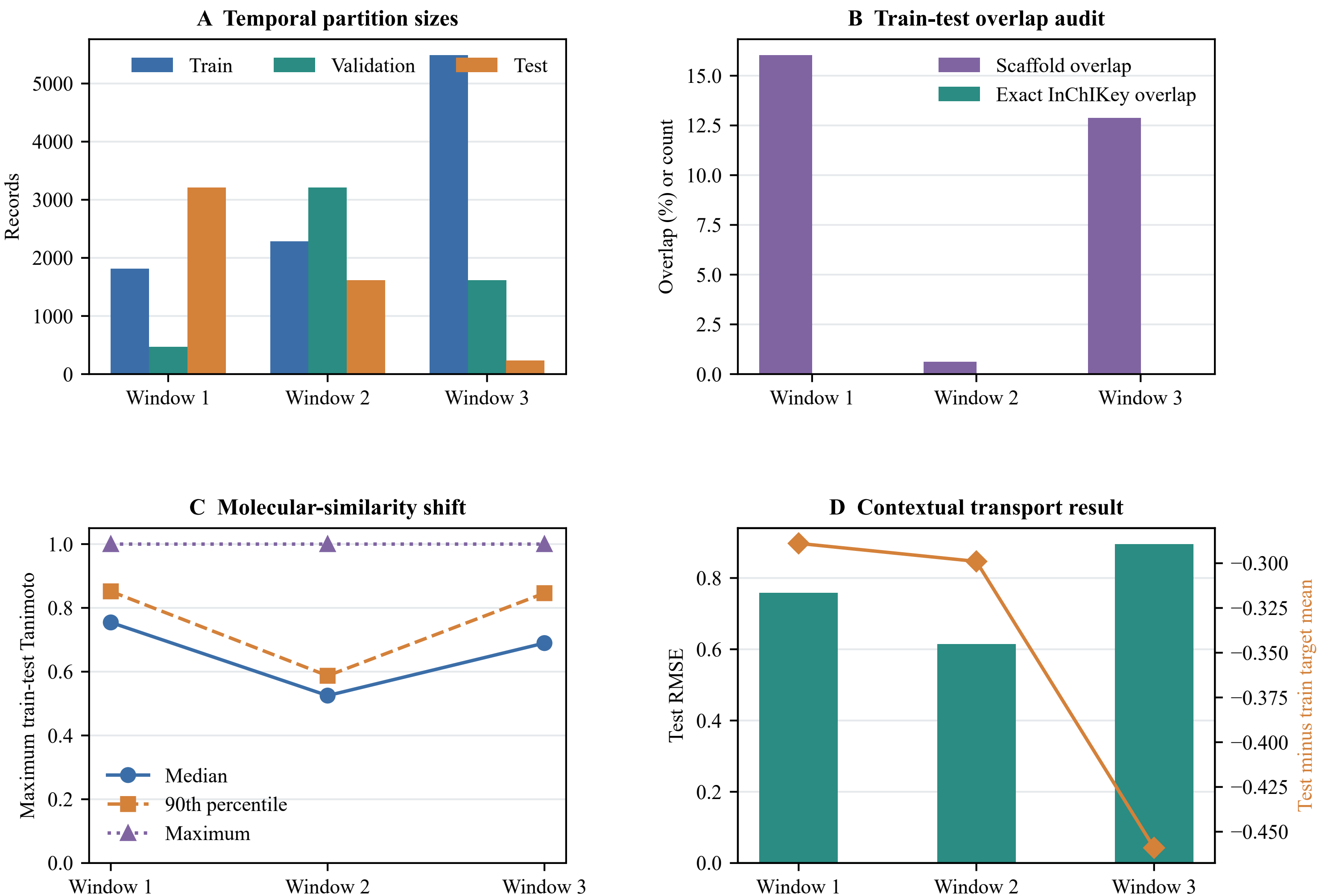
Six endpoints were evaluated at thresholds 0.60, 0.70 and 0.80 using five locked split seeds and the fixed registry. Shading in A–B is the registered-seed bootstrap sensitivity range, not an independent-dataset confidence interval. Only ESOL and Lipophilicity retained the same gap direction at all thresholds; no threshold was chosen after results were inspected.

Supplementary Figure S28.

Independent model-seed variability. Classification ROC-AUC and regression RMSE utilities use separate panels; K=4 is deterministic, whereas larger prefixes frequently changed selected candidate identity.



Supplementary Figure S29. Overlapping forward-temporal contextual stress test



The windows are overlapping sequential stress tests, not independent external cohorts; exact molecular overlap is zero within each train-test window.